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5G PHYSICAL LAYER CHANNEL CONDITIONS CLASSIFICATION USING MACHINE LEARNING TECHNIQUES

Master's Thesis
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ABSTRACT

Long Nguyen: 5G Physical Layer Channel Conditions Classification Using Machine Learning Techniques
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In the field of 5G wireless communication, understanding and classifying the channel environment conditions is crucial for improving transmission performance and enhancing network reliability. Moreover, classification can enhance efficient resource allocation and signal transmission quality, optimizing network performance.

The thesis project concentrates on differentiating the Line-of-Sight (LOS) and Non-Line-of-Sight (NLOS) channel conditions utilizing statistical Tapped Delay Line (TDL) and Clustered Delay Line (CDL) channel models derived from the Sionna framework with data from 5G Physical Uplink Shared Channel (PUSCH). The research incorporates advanced ML techniques for tasks such as data visualization and classification. By harnessing these advanced algorithms, the project seeks to improve the classification accuracy of channel conditions, offering a detailed analysis of their effectiveness in various simulated conditions. The incorporation of Convolutional Neural Network (CNN) is particularly highlighted for its capability to process the complex nature of wireless channel signal data effectively, without relying on traditional feature extraction.

The results of this study demonstrate the efficiency and accuracy of using Deep Learning (DL) compared to supervised ML techniques applied for channel condition detection and classification. Ultimately, the project approach has proven that using signal data as data input without extraction is theoretically possible. The findings are expected to contribute to developing more reliable and efficient 5G communication ML applications.

Keywords: 5G NR, physical layer, data generation, data visualization, machine learning, convolutional neural network

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PREFACE

This thesis was conducted at Nokia from January 2025 to July 2025 for the Computing Sciences master's program at Tampere University. The experiment was performed and supervised by Bo Tan as a Docent of Tampere University and an Associate Professor of University College London, while being advised by Eeli Susan from Nokia.

First and foremost, from the Tampere university perspective, I am deeply grateful to Associate Professor Bo Tan for supervising the thesis and recommending potential approaches for the thesis topic. Moreover, the supplementary communication knowledge and orientation for deepening my theoretical background, while providing a research orientation scheme is worthwhile for me to have a good thesis objective and knowledge for this thesis procedure due to the limitation of a computer science student without a background in communication engineering.

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Next, from the other relational aspect, I appreciate the encouragement from my family and friends who supported me psychologically while doing the thesis.

The last word of thanks is for my girlfriend, Thu Bui. I want to thank you for all the wonderful things you have brought to me while writing this thesis. You are my inspiration, motivation, the person who cares about my sleep and my emotions, making me feel that my efforts, at least until now, always have the mark of your love in them. I hope that we will have the most beautiful ending in the future.

Tampere, Finland, 7th July 2025

Long Nguyen

CONTENTS

1. Introduction 1
   1.1 Thesis Motivation 1
   1.2 Thesis Objective and Methodology Overview 2
   1.3 Thesis Structure 3
2. 5G New Radio 4
   2.1 Overview 4
   2.2 5G NR L1 4
      2.2.1 Frequency Band, Spectrum Flexibility, and Duplex Schemes 4
   2.3 Waveform, Numerology, and Frame Structure 5
   2.4 Physical Resources 7
   2.5 5G Transmission 8
      2.5.1 Channel Mapping Architecture 8
      2.5.2 PUSCH Processing Framework 9
      2.5.3 Demodulation Reference Signal (DMRS) in 5G NR 10
   2.6 5G NR Channel Models 12
      2.6.1 Multipath Channel Propagation 12
      2.6.2 Delay Spread Characteristics 12
      2.6.3 Tapped Delay Line Channel Models 13
      2.6.4 Clustered Delay Line Channel Models 14
3. Statistical Methods for Visualizing High-Dimensional Data 16
   3.1 Importance of Feature Reduction 16
   3.2 t-Distributed Stochastic Neighbor Embedding 17
4. Deep Learning 18
   4.1 Artificial Neural Network 18
      4.1.1 Perceptron: The Fundamental Unit 18
      4.1.2 Feedforward Neural Network Architecture 19
   4.2 Convolutional Neural Network (CNN) 21
5. Methods 24
   5.1 Research Methodology Overview 24
   5.2 Sionna 25
      5.2.1 Introduction 25
      5.2.2 Features 26
   5.3 Dataset Generation Methodology 26
      5.3.1 Transmitter Processing 27

5.3.2 Channel Model Application 27
5.3.3 Label Generation Methodology 27
5.3.4 Receiver Signal Capture 28
5.4 Simulation Parameters Configuration 28
5.4.1 5G NR PUSCH Parameters 28
5.4.2 Channel Model and Antenna Configuration 29
5.4.3 Simulation Parameters 30
5.5 Data visualization 31
5.5.1 Received symbols. 31
5.5.2 Data statistical visualization 33
5.6 Neural Network Architecture 36
5.6.1 Multilayer Perceptron Model 36
5.6.2 VGGNet and ResNet Architectural Principles 36
5.6.3 Complex-Valued OFDM Channel Processing 38
5.6.4 VGGNet Based Model Architecture Overview 38
5.6.5 ResNet Based Model Architecture Overview 40
5.6.6 Task-Specific Architecture Details 42
6. Results 44
6.1 LOS/NLOS Classification Results 44
6.1.1 Loss and Accuracy Analysis for NLOS/LOS classification 44
6.1.2 Evaluation Metrics Results for NLOS/LOS Classification 46
6.1.3 SNR-Dependent Performance for NLOS/LOS Classification 48
6.2 Channel Model Classification Results 49
6.2.1 Loss and Accuracy Analysis for Channel Classification 49
6.2.2 Evaluation Metrics Results for Channel Classification 51
6.2.3 SNR-Dependent Performance in Channel Classification 53
7. Conclusion 56
7.1 Summary of Findings 56
7.2 Future Research Directions. 56
References 58
Appendix A: TDL Channel Profile 62
Appendix B: CDL Channel Profile 69

LIST OF FIGURES

2.1 5G NR frame structure [16] 6
2.2 5G NR Physical Resources Structure 7
2.3 Uplink channel mapping hierarchy in 5G NR showing the relationship between PUSCH and other physical channels [18] 9
2.4 DMRS mapping configurations showing (a) Type A front-loaded structure and (b) Type B distributed pattern. Figure originated from [20] 11
2.5 DMRS configuration examples for different parameter combinations showing (a) basic configuration, (b) extended length, and (c) additional positions [20]. 11
2.6 Relationship between CDL and TDL models showing how spatial filtering and antenna patterns transform the full spatial channel model into a simplified tapped delay line representation [19] 14

4.1 Complete structure of an artificial neuron with activation function 19
4.2 A FNN structure is demonstrated. 20
4.3 Filtering operation is depicted in a convolutional layer, adapted from [31]. . 22

5.1 Research workflow with two processing paths: Feature extraction for traditional ML and visualization, Direct neural network processing. 24
5.2 Constellation diagrams of 1 sample for each TDL channel model A-E. . . . 31
5.3 Constellation diagrams of 1 sample for each CDL channel model A-E. . . . 32
5.4 t-SNE projection showing natural clustering of LOS (blue) and NLOS (red) conditions across all channel models. The visualization suggests reasonable separability between propagation conditions. 34
5.5 t-SNE projection showing clustering patterns for all ten channel models (TDL A-E and CDL A-E). Colors represent different model variants, demonstrating limited model-specific separation. 35
5.6 Residual block architecture demonstrating the skip connection that enables identity mapping. The direct path preserves signal integrity while allowing gradients to flow unimpeded through deep networks. 37
5.7 BasicBlock demonstration of block $12 \times 7 \times 128$ of the Table 5.7 41

6.1 VGGNet (SGD) training losses and accuracy for LOS/NLOS classification. 44
6.2 ResNet (Adam) training losses and accuracy for LOS/NLOS classification. 45

6.3 Comparative confusion matrices for NLOS/LOS classification showing (a) VGGNet SISO, (b) VGGNet MIMO, (c) ResNet SISO, and (d) ResNet MIMO configurations. 47
6.4 VGGNet channel classification training and accuracy 50
6.5 ResNet channel classification metrics and accuracy. 50
6.6 Channel model classification confusion matrices showing (a) VGGNet SISO, (b) VGGNet MIMO, (c) ResNet SISO, and (d) ResNet MIMO configurations. 52

LIST OF TABLES

2.1 5G NR Numerology Parameters [17] 6
2.2 PUSCH Modulation Schemes 9
2.3 Example Delay Spread Parameters for TDL/CDL Channel Models [19] . . . 13

5.1 Key Components in Sionna Framework [35] 26
5.2 5G NR PUSCH System Configuration 29
5.3 Channel and Antenna Configuration 30
5.4 Simulation Control Parameters 30
5.5 MLP Training Configuration 36
5.6 Adaptive VGGNet Architecture for OFDM Processing: SISO (1×1 conv) and MIMO (3×3 conv) configurations 39
5.7 ResNet Architecture for OFDM Processing 40
5.8 Architecture Parameters by Task 43

6.1 NLOS/LOS Classification Performance Metrics Comparison 47
6.2 Algorithm Configurations for NLOS/LOS Classification 48
6.3 NLOS/LOS Classification Accuracy Across SNR Conditions (%) 49
6.4 Channel Model Classification Performance Metrics Comparison 52
6.5 Channel Classification Accuracy Across SNR Conditions (%) 54

A.1 TDL-A Channel Model Parameters (NLOS) 63
A.2 TDL-B Channel Model Parameters (NLOS) 64
A.3 TDL-C Channel Model Parameters (NLOS) 65
A.4 TDL-D Channel Model Parameters (LOS) 66
A.5 TDL-E Channel Model Parameters (LOS) 67
A.6 TDL-E Channel Model Parameters (LOS) 68

B.1 CDL-B Channel Model Parameters (NLOS) 69
B.2 CDL-A Channel Model Parameters (NLOS) 70
B.3 CDL-C Channel Model Parameters (NLOS) 71
B.4 CDL-D Channel Model Parameters (LOS) 72
B.5 CDL-E Channel Model Parameters (LOS) 73

LIST OF SYMBOLS AND ABBREVIATIONS

16QAM	16-point Quadrature Amplitude Modulation
1G	First generation
256QAM	256-point Quadrature Amplitude Modulation
3GPP	3rd Generation Partnership Project
4G LTE	Fourth generation Long-Term Evolution
5G	Fifth generation
5G NR	Fifth generation New Radio
64QAM	64-point Quadrature Amplitude Modulation
Adam	Adaptive Moment Estimation
AI	Artificial Intelligence
ANN	Artificial Neural Networks
AWGN	Additive White Gaussian Noise
BS	Base Station
BWP	Bandwidth Part
CDL	Clustered Delay Line
CDM	Code-Division Multiplexing
CNN	Convolutional Neural Network
CP-OFDM	Cyclic Prefix Orthogonal Frequency Division Multiplexing
CSI	Channel state information
DFT-S-OFDM	Discrete Fourier Transform Spread Orthogonal Frequency Division Multiplexing
DL	Deep Learning
DMRS	Demodulation Reference Signal
DS	Delay Spread
eMBB	Enhanced mobile broadband
FDD	Frequency Division Duplex Bands
FNN	Feed-Forward Neural Network

IoT	Internet of Things
ISI	Inter-symbol Interference
kNN	k-Nearest Neighbor
L1	Physical Layer
LOS	Line-of-Sight
MAC	Medium access control
MIMO	Multiple-Input Multiple-Output
ML	Machine Learning
MLP	Multilayer Perceptron
MMSE	Minimum Mean Square Error
NLOS	Non-Line-of-Sight
NN	Neural Network
OFDM	Orthogonal Frequency Division Multiplexing
PCA	Principal Component Analysis
PDSCH	Physical Downlink Shared Channel
PRB	Physical Resource Blocks
PT-RS	Phase-Tracking Reference Signal
PUSCH	Physical Uplink Shared Channel
QPSK	Quadrature Phase-Shift Keying
RE	Resource Element
RMS	Root Mean Square
RNTI	Radio Network Temporary Identifier
SCS	subcarrier spacing
SDL	Supplementary Downlink
SGD	Stochastic Gradient Descent
SISO	Single-Input Single-Output
SNR	Signal-to-Noise Ratio
SUL	Supplementary Uplink
SVM	Support Vector Machine
t-SNE	t-distributed Stochastic Neighbor Embedding
TDD	Time Division Duplex Bands
TDL	Tapped Delay Line

UE	User Equipment
URLLC	Ultra-reliable low-latency communications
UWB	Ultra-Wideband
VRB	Virtual Resource Block

1. INTRODUCTION

1.1 Thesis Motivation

The evolution of mobile communication over the past four decades, commencing with the introduction of First generation (1G) networks in 1979, has culminated in the development of Fifth generation (5G) mobile communication networks. These networks promise to transform speed, capacity, and connectivity. Fifth generation New Radio (5G NR), as the latest standard succeeding Fourth generation Long-Term Evolution (4G LTE), is specifically designed to address the escalating demands for high data rates, low latency, and massive device connectivity.

Recent advancements in computational technology have been instrumental in developing and deploying Machine Learning (ML) models across various domains [1], including wireless communications. Within this context, ML has emerged as a powerful paradigm for enhancing 5G NR systems, offering data-driven solutions to fundamental wireless communication challenges. In contrast to traditional model-based approaches, ML techniques can autonomously extract latent patterns from channel measurements, enabling intelligent adaptation to dynamic environments. This capability proves particularly valuable in challenging communication scenarios characterized by high mobility and severe multipath fading [2]. These developments create novel opportunities to optimize 5G NR systems beyond the limitations of conventional signal processing techniques.

The realization of 5G NR's ambitious capabilities necessitates efficient physical layer processing at the 5G NR Physical Layer (L1) level. This requirement motivates the development of ML-based channel condition classification methods capable of directly distinguishing between different channel states using unprocessed PUSCH data. Such methods could provide valuable environmental insights based on readily available input signals. Among well-established channel condition concepts, the classification of Non-Line-of-Sight (NLOS) and Line-of-Sight (LOS) scenarios has been extensively researched in Ultra-Wideband (UWB) [3, 4, 5] and Wi-Fi [6, 7] applications. Regarding 5G NR research, this classification subject finds numerous applications, including localization within mobile communication networks and network performance optimization through environment-aware resource allocation algorithms [8].

Existing NLOS/LOS classification techniques in UWB and Wi-Fi systems predominantly rely on channel impulse response analysis and handcrafted feature extraction, methods well-suited to their constrained frequency bands and specific use cases. However, 5G networks present unique challenges due to their expansive frequency spectrum and advanced beamforming capabilities, necessitating more sophisticated classification approaches. While prior research has investigated supervised ML methods for 5G Multiple-Input Multiple-Output (MIMO) systems and Physical Downlink Shared Channel (PDSCH)-assisted Deep Learning (DL) classification [9, 10, 11], the potential of Physical Uplink Shared Channel (PUSCH) signals—a universally available resource in 5G baseband processing—remains largely unexplored. This thesis addresses this research gap by proposing novel ML-based solutions for channel condition classification using unprocessed PUSCH data, offering a practical and scalable alternative to existing methodologies.

1.2 Thesis Objective and Methodology Overview

This thesis aims to develop an efficient ML-based framework for classifying LOS and NLOS channel conditions in 5G NR systems using synthetic PUSCH signals. An additional objective is to classify different channel model environments. The research employs various ML algorithms as baseline classifiers while exploring more advanced approaches.

The primary objectives and methodological approach encompass several key aspects. First, comprehensive channel datasets will be generated through Tapped Delay Line (TDL) and Clustered Delay Line (CDL) models adhering to 3rd Generation Partnership Project (3GPP) standards.

Second, the research incorporates data visualization and validation techniques. Since the data originates from Sionna simulations, t-distributed Stochastic Neighbor Embedding (t-SNE) will be applied to visualize data distributions and assess the statistical separability of different channel conditions. This analysis provides foundational insights into the dataset's characteristics and classification potential.

Third, the study focuses on designing and optimizing ML-based models. Initial investigations employ supervised ML techniques including k-Nearest Neighbor (kNN), Support Vector Machine (SVM), Random Forests, and Multilayer Perceptron (MLP) to establish performance baselines and reveal the limitations of traditional approaches. Subsequently, more sophisticated Neural Network (NN) architectures, particularly Convolutional Neural Network (CNN), will be developed and evaluated for classification purposes.

Finally, comprehensive result analysis will be conducted using established evaluation metrics to assess model performance and robustness. The anticipated contribution of this research lies in advancing both the 5G NR field and DL applications in mobile communi-

cation networks.

1.3 Thesis Structure

This thesis is systematically organized into seven chapters that progress from theoretical foundations to experimental validation. Following this introductory chapter, Chapter 2 establishes the theoretical framework of 5G NR systems, with particular emphasis on L1 processing and computational methods for wireless communications. Chapter 3 investigates statistical approaches for visualizing high-dimensional channel data, including dimensionality reduction techniques critical for analyzing propagation characteristics.

The DL paradigm is presented in Chapter 4, which covers fundamental architectures and their adaptations for wireless signal processing. Chapter 5 details the methodological framework, encompassing data generation through channel models, model development. Comprehensive experimental results and comparative analyses between CNN and conventional machine learning approaches are presented in Chapter 6. Finally, Chapter 7 synthesizes the key findings, discusses their implications for 5G NR systems, and outlines promising directions for future research in ML-enhanced wireless communications.

2. 5G NEW RADIO

2.1 Overview

The 5G NR standard, introduced in 3GPP Releases 16 and 17, represents a transformative leap in wireless communication by addressing the limitations of 4G LTE in scalability, latency, and spectral efficiency. Unlike its predecessors, 5G NR supports heterogeneous use cases through three primary service categories: eMBB for high-throughput applications such as 4K video streaming, Ultra-reliable low-latency communications (URLLC) for mission-critical systems like industrial automation, and mMTC (massive machine-type communications) for IoT deployments [12]. Key innovations include support for millimeter-wave (mmWave) frequencies, massive MIMO, and flexible numerology, enabling peak data rates exceeding 20 Gbps and latencies below 1 ms. The architecture is divided into two frequency ranges: FR1 (sub-6 GHz) for wide-area coverage and FR2 (mmWave) for ultra-high capacity, with network slicing allowing operators to partition resources dynamically for diverse applications. Recent developments have also proposed FR3 (7.125–24 GHz) as an upper mid-band extension to bridge the gap between FR1 and FR2, though this remains under study in 3GPP Release 19. Network slicing allows operators to dynamically partition resources for diverse applications across these frequency ranges.

2.2 5G NR L1

2.2.1 Frequency Band, Spectrum Flexibility, and Duplex Schemes

The development of each mobile network connection evolution leads to the expansion of bands at different frequencies. Higher frequencies (mmWave) provide greater bandwidth for enhanced throughput but suffer from limited propagation range, making them suitable only for dense environments, while lower frequencies offer wider coverage with more modest capacity gains [13]. 5G NR Physical Layer (L1) operates across two primary frequency ranges:

- Frequency Range 1 (FR1) (450 MHz–6 GHz): often termed "sub-6 GHz," provides wide-area coverage and supports legacy compatibility through dynamic spectrum sharing (DSS), enabling simultaneous 4G and 5G transmissions on the same carrier.

- Frequency Range 2 (FR2) (24.25–71 GHz): often known as "mmWave" divided into two sub-ranges - FR2-1 (24.25-52.6 GHz) and FR2-2 (52.6-71 GHz)

The deployment of FR1 and FR2 shapes their suitability for different use cases. FR1 offers superior coverage and penetration, making it ideal for wide-area deployments in urban, suburban, and rural environments. Its compatibility with existing 4G infrastructure allows for cost-effective network upgrades, while its balanced uplink/downlink performance supports applications like mobile broadband and Internet of Things (IoT). In contrast, FR2 delivers ultra-high throughput and ultra-low latency due to its vast available bandwidth, enabling multi-Gbps speeds for dense urban hotspots, fixed wireless access, and immersive technologies like AR/VR. The use of advanced beamforming in FR2 mitigates some propagation challenges, though its effectiveness remains limited to short-range, mainly LOS scenarios. In realistic scenarios, different operating bands are used with different downlink and uplink ranges in different regions and duplex modes [14]. 5G NR can utilize Frequency Division Duplex Bands (FDD) and Time Division Duplex Bands (TDD), with the additional duplexing such as Supplementary bands (Supplementary Uplink (SUL) and Supplementary Downlink (SDL)) or applying Unlicensed Spectrum.

2.3 Waveform, Numerology, and Frame Structure

In 5G, the main applied waveform is Orthogonal Frequency Division Multiplexing (OFDM), which is a multi-carrier modulation technique that divides a high-rate data stream into multiple parallel subcarriers, enabling efficient spectrum utilization and robustness against frequency-selective fading.5G NR adopts Cyclic Prefix Orthogonal Frequency Division Multiplexing (CP-OFDM) for both downlink and uplink transmissions, ensuring compatibility with multi-antenna systems because the added cyclic prefix mitigates Inter-symbol Interference (ISI) and maintains orthogonality in multipath environments, making it more robust for high-speed wireless communications [15]. meanwhile, for uplink scenarios requiring low SNR case, such as IoT devices with limited power budgets, Discrete Fourier Transform Spread Orthogonal Frequency Division Multiplexing (DFT-S-OFDM) is optionally employed [16]. In this thesis application, OFDM will be used as the fundamental waveform in Single-Input Single-Output (SISO) and Multiple-Input Multiple-Output (MIMO) environments due to its well-established performance in handling multipath fading, its compatibility with advanced multi-antenna techniques, and its widespread adoption in modern wireless standards, which simplifies implementation and validation.

The performance of OFDM in 5G NR is further optimized through an adaptive numerology framework. This system supports five distinct numerologies ($\mu \in \{0, 1, 2, 3, 4\}$), with subcarrier spacing (SCS) scaling according to:

$$\Delta f = 2^{\mu} \times 15\,\text{kHz} \tag{2.1}$$

where μ represents the numerology index. This flexible approach enables 5G NR to accommodate diverse service requirements, ranging from eMBB to URLLC.

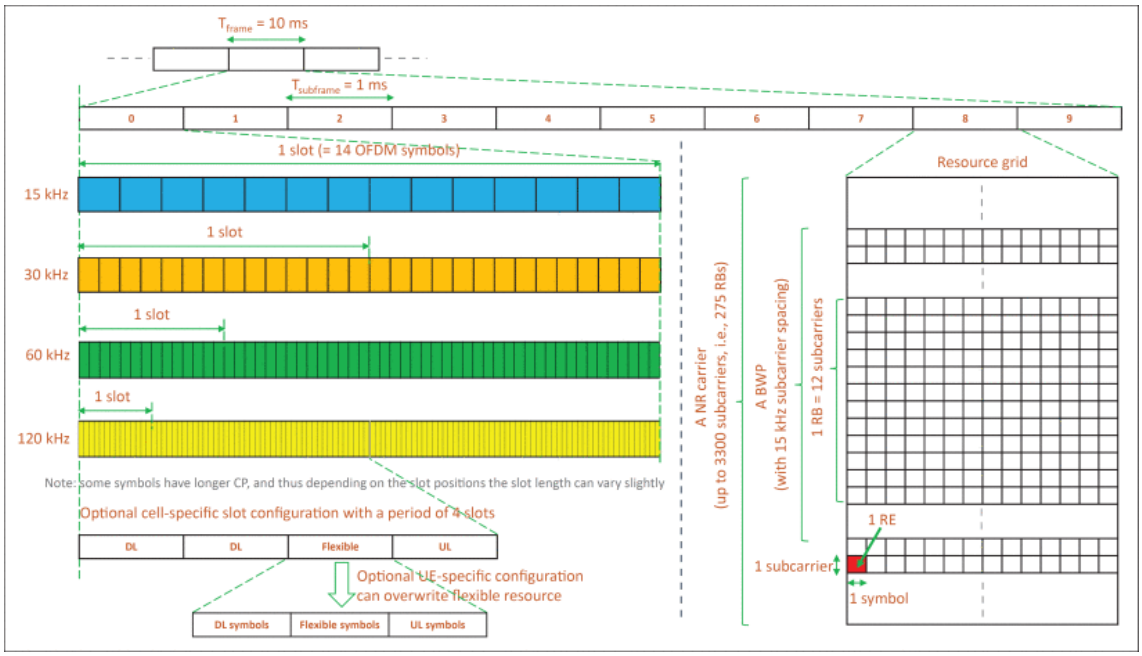


Figure 2.1. *5G NR frame structure [16]*

The choice of numerology could be observed in Figure 2.1. The 5G NR numerology framework establishes a direct relationship between SCS and slot duration, as illustrated in Figure 2.1. The fundamental design follows an inverse proportional relationship where doubling the SCS ($\Delta f = 2^{\mu} \times 15$ kHz) exactly halves the slot duration ($T_{slot} = 1/2^{\mu}$ ms). This scaling ensures temporal consistency across numerologies while maintaining orthogonality - a 15 kHz configuration ($\mu = 0$) provides 1 ms slots suitable for wide-area coverage, while 120 kHz ($\mu = 3$) yields 0.125 ms slots for latency-critical applications. The slot duration reduction comes at the cost of increased overhead from more frequent cyclic prefix insertion, but enables finer time-domain granularity for URLLC services.

Table 2.1. *5G NR Numerology Parameters [17]*

Numerology (μ)	SCS (kHz)	Slot Length (ms)	Max BW (MHz)
0	15	1.0	49.5
1	30	0.5	99
2	60	0.25	198
3	120	0.125	396
4	240	0.0625	-

As shown in Table 2.1, higher numerologies enable wider channel bandwidths (up to 400

MHz in FR2) by reducing phase noise sensitivity through shorter symbols.

2.4 Physical Resources

The 5G NR physical layer structure organizes transmission resources in both time and frequency domains. In the time dimension, OFDM symbols are grouped into slots (14 symbols each), which combine to form subframes (1 ms) and ultimately complete radio frames (10 ms). In frequency, subcarriers are bundled into Physical Resource Blocks (PRB)s, creating a two-dimensional resource grid that serves as the foundation for all transmissions. In general, 5G physical resources can be represented in Figure 2.2 below:

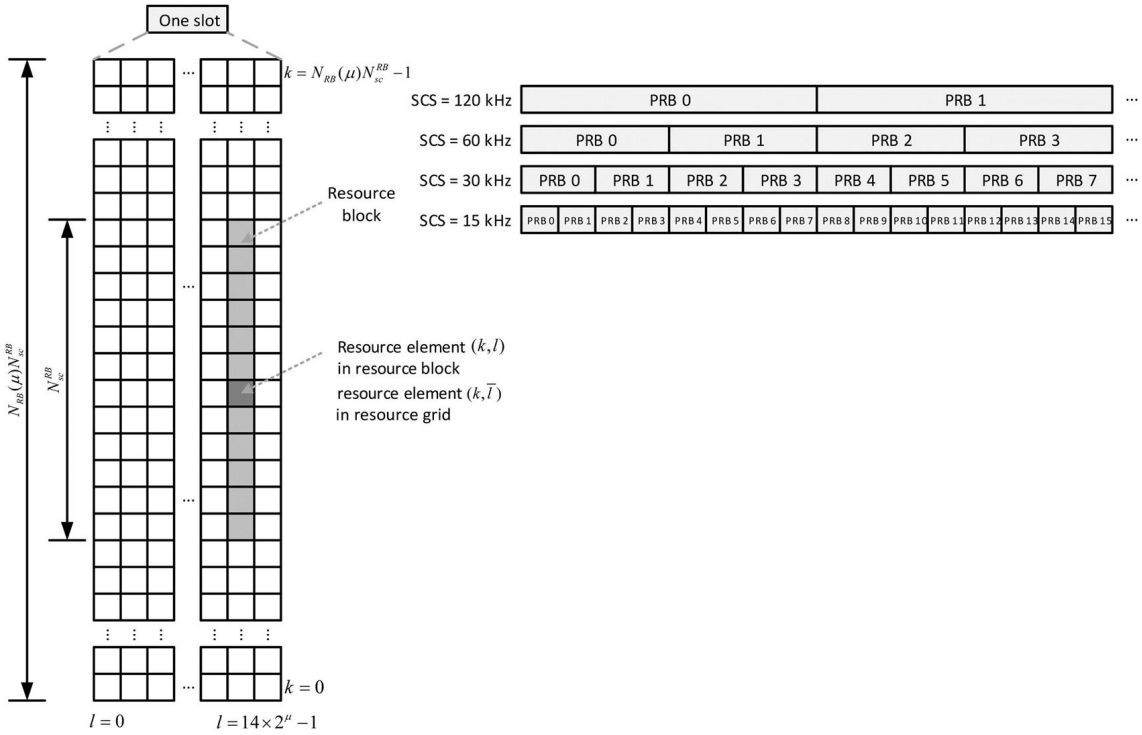


Figure 2.2. *5G NR Physical Resources Structure*

The complete physical resource architecture consists of four fundamental components, which could be observed in the Figure 2.2:

Resource Grid: This time-frequency matrix forms the basic transmission canvas, with its dimensions adapting to the selected numerology. Each grid configuration is specific to a particular subcarrier spacing (15-240 kHz), antenna port configuration, and transmission direction (uplink/downlink). The grid's horizontal axis represents the available subcarriers within the active bandwidth, while the vertical axis corresponds to the OFDM symbols within a transmission interval.

Resource Element (RE): As the smallest addressable unit, an RE represents the intersection of one subcarrier (frequency domain) and one OFDM symbol (time domain).

While the absolute duration and spacing vary with numerology, each RE always maintains this fundamental 1:1 relationship between time and frequency components. Data symbols, reference signals, and control information are all mapped to specific REs within the grid.

Physical Resource Blocks (PRB): The standard allocation unit consists of 12 consecutive subcarriers spanning one complete slot (typically 14 symbols). This 12×14 RE structure (168 total REs for normal cyclic prefix) remains consistent across numerologies, though the physical dimensions in time and frequency scale with the subcarrier spacing. PRBs enable efficient resource allocation while maintaining compatibility with various bandwidth configurations.

Bandwidth Part (BWP): A flexible carrier segment comprising contiguous PRBs, BWPs allow networks to support multiple services simultaneously. Key characteristics include

- Configurable bandwidth (24-275 PRBs depending on SCS)
- Independent numerology per BWP
- Dynamic activation/switching
- UE-specific bandwidth reduction

This hierarchical structure enables 5G NR to efficiently support diverse services while optimizing resource utilization across varying channel conditions and device capabilities. The physical layer precisely maps information streams onto these time-frequency resources through sophisticated scheduling algorithms that account for channel quality, traffic requirements, and interference conditions.

2.5 5G Transmission

2.5.1 Channel Mapping Architecture

The 5G NR transmission system organizes physical channels through a structured mapping process, as illustrated in Figure 2.3. The Physical Uplink Shared Channel (PUSCH) serves as the primary uplink data channel, responsible for carrying UE data and higher-layer signaling while supporting dynamic bandwidth adaptation and MIMO transmission. Its position in the protocol stack bridges the MAC layer's transport blocks to the physical layer's REs through a carefully designed signal processing chain, as shown in Figure 2.3.

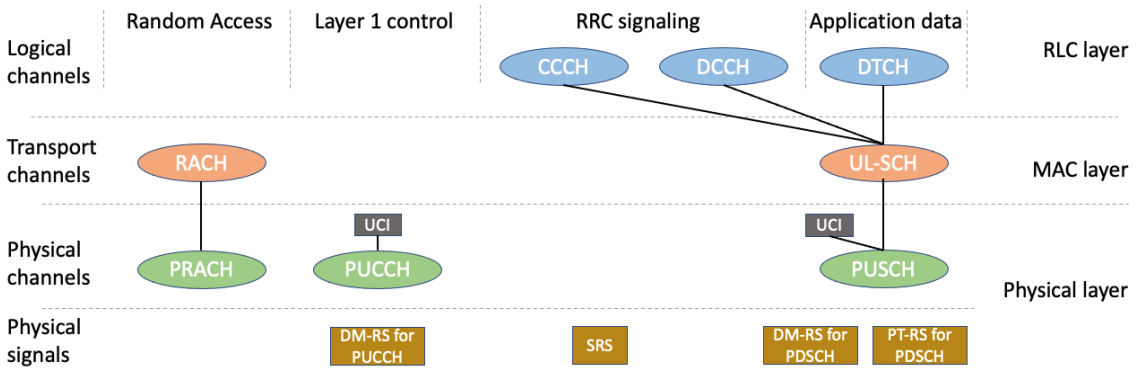


Figure 2.3. *Uplink channel mapping hierarchy in 5G NR showing the relationship between PUSCH and other physical channels [18]*

2.5.2 PUSCH Processing Framework

The PUSCH transmission undergoes seven critical processing stages:

1. **Scrambling**: Applies UE-specific pseudorandom sequences using a length-31 Gold sequence generator initialized with CSI and RNTI parameters. This randomization reduces inter-cell interference and enhances physical-layer security.

2. **Modulation**: Converts scrambled bits to complex-valued symbols using adaptive modulation schemes. Table 2.2 presents the supported orders:

Table 2.2. *PUSCH Modulation Schemes*

Transform Precoding Enabled	**Scheme**
1	$\pi/2$-BPSK (DFT-S-OFDM)
2	QPSK
4	16QAM
6	64QAM
8	256QAM
Transform Precoding Disabled	**Scheme**
2	QPSK
4	16QAM
6	64QAM
8	256QAM

3. **Layer Mapping**: Distributes symbols across multiple antenna ports (up to 4 layers) using a frequency-first approach to maintain contiguity and enable MIMO operation.

4. **Transform Precoding**: Optional stage for DFT-S-OFDM that applies:

$$x_l(n) = \frac{1}{\sqrt{N_{\text{PRB}}}} \sum_{k=0}^{N_{\text{PRB}}-1} d_l(k) e^{-j2\pi nk/N_{\text{PRB}}}, \quad 0 \leq n \leq N_{\text{PRB}} - 1 \tag{2.2}$$

where N_{PRB} is the number of allocated PRBs, $d_l(k)$ are frequency-domain symbols for layer l, and $x_l(n)$ are time-domain samples.

5. **Precoding**: Applies antenna-specific weights from configured codebooks based on CSI to optimize radiation patterns.

6. **Virtual Resource Mapping**: Assigns symbols to either interleaved or localized Virtual Resource Block (VRB)s for frequency diversity or scheduling flexibility.

7. **Physical Resource Mapping**: Translates VRBs to PRBs within the active BWP, incorporating:

- Demodulation Reference Signal (DMRS) symbol positions
- Phase-Tracking Reference Signal (PT-RS) for phase tracking
- Frequency-domain allocation types

This comprehensive processing enables PUSCH to support diverse 5G services while maintaining waveform flexibility and efficient resource utilization.

2.5.3 Demodulation Reference Signal (DMRS) in 5G NR

The Demodulation Reference Signal (DMRS) constitutes a fundamental reference signal in 5G NR systems, enabling coherent demodulation of both uplink (PUSCH) and downlink (PDSCH) transmissions. As specified in [17], this UE-specific design represents a significant evolution from 4G LTE's cell-specific reference signals, offering enhanced support for MIMO operations and adaptability to diverse channel conditions.

The DMRS configuration provides two distinct mapping types, as illustrated in Figure 2.4. Type A employs front-loaded symbols (positions 2-3 within the slot) specifically optimized for static or low-mobility scenarios, delivering low-latency channel estimation with minimal overhead. This configuration proves particularly effective in LOS conditions and for eMBB applications [17]. Conversely, Type B utilizes a distributed pattern across symbols 0, 4, 7, and 11, demonstrating superior performance in high Doppler scenarios where rapid channel variations must be tracked, making it ideal for NLOS environments and URLLC requirements [19].

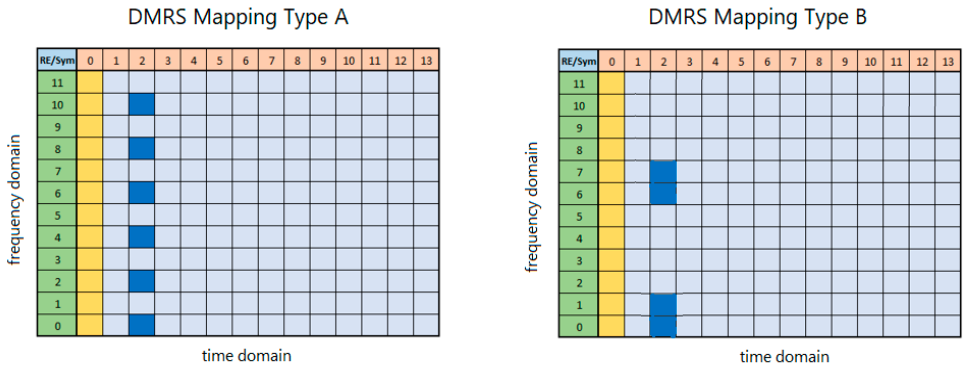


Figure 2.4. *DMRS mapping configurations showing (a) Type A front-loaded structure and (b) Type B distributed pattern. Figure originated from [20]*

Figure 2.5 demonstrates the practical implementation of DMRS through various parameter combinations. The standard supports flexible configurations through three primary parameters: the `dmrs-Type` parameter (values 1 or 2) determines the sequence generation method, while `dmrs-maxLength` (1 or 2) controls whether single or double symbol duration is used. The `dmrs-AdditionalPosition` parameter (0-3) specifies the number of additional DMRS symbols, allowing for adaptation to different channel conditions and quality requirements.

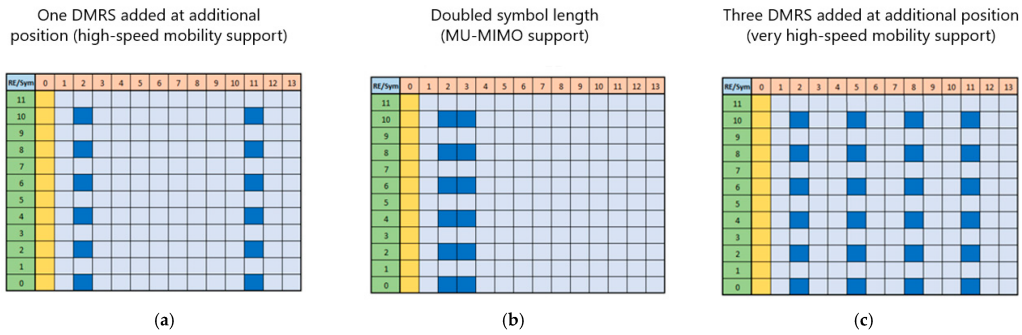


Figure 2.5. *DMRS configuration examples for different parameter combinations showing (a) basic configuration, (b) extended length, and (c) additional positions [20].*

DMRS will be used for channel estimation. The Minimum Mean Square Error (MMSE) estimation approach forms the foundation for DMRS processing, expressed as:

$$\hat{\mathbf{H}}_{\text{MMSE}}[p] = \mathbf{R}_{\mathbf{HH}} \left(\mathbf{R}_{\mathbf{HH}} + \beta^{-1} \mathbf{\Lambda}^{-1} \right)^{-1} \hat{\mathbf{H}}_{\text{LS}}[p] \tag{2.3}$$

where $\hat{\mathbf{H}}_{\text{MMSE}}[p]$ represents the MMSE channel estimate at DMRS symbol position p, $\mathbf{R}_{\mathbf{HH}} = \mathbb{E}\{\mathbf{H}\mathbf{H}^H\}$ denotes the channel correlation matrix in frequency domain, and β corresponds to the estimated signal-to-noise ratio. The diagonal matrix $\mathbf{\Lambda}$ accounts for the DMRS power allocation across subcarriers, while $\hat{\mathbf{H}}_{\text{LS}}[p] = \mathbf{Y}[p] \oslash \mathbf{X}_{\text{DMRS}}[p]$ gives the

least-squares estimate with $\oslash$ indicating element-wise division. Advanced implementations increasingly incorporate ML techniques, particularly for challenging scenarios such as mmWave channel estimation, compensation of non-linear effects, and improved tracking of DS in wideband systems [21].

2.6 5G NR Channel Models

2.6.1 Multipath Channel Propagation

The wireless propagation environment in 5G networks exhibits complex multipath characteristics that significantly impact system performance. When a transmitted signal encounters obstacles such as buildings, vehicles, or terrain features, it undergoes reflection, diffraction, and scattering, resulting in multiple delayed versions of the original signal arriving at the receiver. This multipath propagation creates time-varying channel conditions characterized by frequency-selective fading and ISI. The baseband channel impulse response for such environments can be mathematically represented as:

$$\mathbf{H}(\tau, t) = \sum_{n=1}^{N_p(t)} \alpha_n(t) e^{j\theta_n(t)} \delta(\tau - \tau_n(t)) \tag{2.4}$$

where $N_p(t)$ is the time-varying number of multipath components, $\alpha_n(t)$ is the n-th path amplitude, $\theta_n(t)$ is the phase shift, and $\tau_n(t)$ is the time delay. The Doppler effect caused by relative motion between transmitter and receiver introduces additional frequency shifts to each multipath component, given by $f_{D,n} = \frac{v}{\lambda} \cos \phi_n$, where v is the relative velocity, λ is the wavelength, and ϕ_n is the arrival angle of the n-th path.

2.6.2 Delay Spread Characteristics

The Delay Spread (DS) is a fundamental parameter characterizing multipath channels, defined as the time dispersion between the first and last significant signal components arriving at the receiver. The Root Mean Square (RMS) DS provides a quantitative measure:

$$\tau_{rms} = \sqrt{\frac{\sum_k P_k \tau_k^2}{\sum_k P_k} - \left(\frac{\sum_k P_k \tau_k}{\sum_k P_k} \right)^2} \tag{2.5}$$

where P_k is the power and τ_k is the delay of the k-th path. This parameter critically determines the channel's frequency selectivity - when the symbol duration is smaller than τ_{rms}, the channel exhibits frequency-selective fading.

For system evaluation, typical DS scenarios for TDL and CDL are categorized as:

Table 2.3. *Example Delay Spread Parameters for TDL/CDL Channel Models [19]*

Model Type	**Desired Delay Spread (DS)**
Very short delay spread	10 ns
Short delay spread	30 ns
Nominal delay spread	100 ns
Long delay spread	300 ns
Very long delay spread	1000 ns

These values serve as example scaling parameters and can be adjusted to capture specific scenarios or beamforming effects. The actual experienced DSs may vary significantly depending on the deployment environment, ranging from nanoseconds in indoor scenarios to microseconds in urban macrocell environments.

2.6.3 Tapped Delay Line Channel Models

The TDL model provides a discrete-time representation of multipath channels by approximating the continuous channel impulse response with a finite number of time delayed taps. Each tap corresponds to a cluster of multipath components with similar delays. The 3GPP specification defines TDL models that can be configured with different DS characteristics to represent various propagation conditions, in which the channel parameters are defined in Appendix A.

The TDL model provides a discrete-time representation of multipath channels through its channel impulse response:

$$\mathbf{H}(t,\tau) = \sum_{k=1}^{K} c_k(t)\delta(\tau - \tau_k) \tag{2.6}$$

where K is the number of channel taps, $c_k(t)$ represents the complex time-varying coefficient of the k-th tap, and τ_k is the corresponding tap delay. Each tap's coefficient $c_k(t)$ follows a Rayleigh or Rician distribution depending on the presence of a dominant LOS component, with temporal variations modeled by the Jakes' autocorrelation function:

$$R_{c_k}(\Delta t) = J_0(2\pi f_{D,k}\Delta t) \tag{2.7}$$

where $J_0(\cdot)$ is the zero-order Bessel function of the first kind and $f_{D,k}$ is the maximum Doppler frequency for the k-th tap.

2.6.4 Clustered Delay Line Channel Models

For more accurate modeling of millimeter-wave and MIMO channels, 5G NR employs Clustered Delay Line (CDL) models that capture both spatial and temporal characteristics. As shown in Figure 2.6, CDL models incorporate spatial filtering and antenna pattern effects that are abstracted in TDL representations. These models organize multipath components into clusters, where each cluster contains several subpaths with specific angular properties.

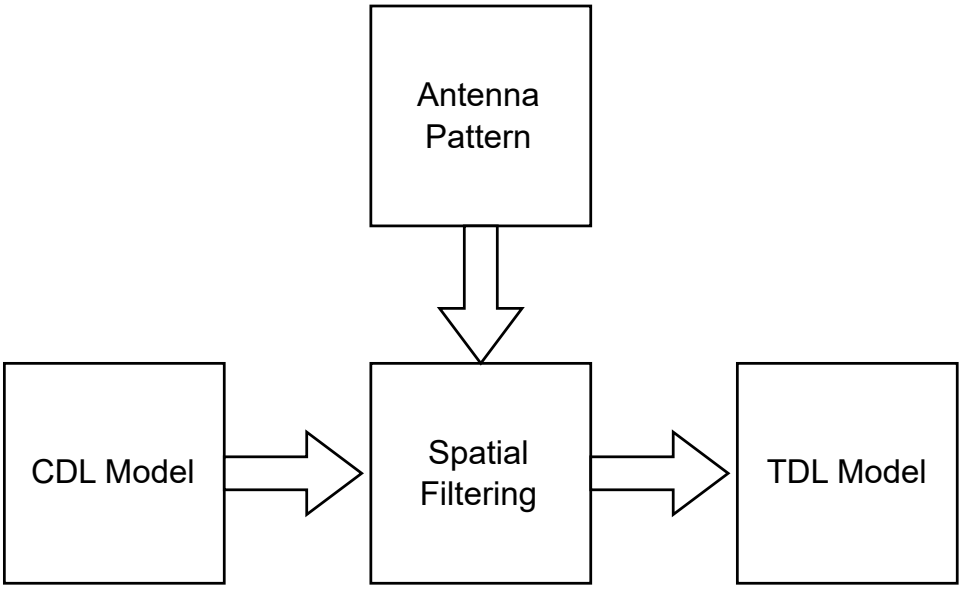


Figure 2.6. *Relationship between CDL and TDL models showing how spatial filtering and antenna patterns transform the full spatial channel model into a simplified tapped delay line representation [19]*

The CDL channel impulse response extends the basic multipath model to include spatial dimensions:

$$\mathbf{H}(t,\tau) = \sum_{c=1}^{N_c} \sum_{m=1}^{M_c} \alpha_{c,m}(t)\mathbf{a}_r(\theta_{c,m}^r, \phi_{c,m}^r)\mathbf{a}_t^H(\theta_{c,m}^t, \phi_{c,m}^t)\delta(\tau - \tau_c) \tag{2.8}$$

where N_c represents the number of clusters, M_c denotes the number of subpaths in the c-th cluster, $\alpha_{c,m}(t)$ is the complex gain of the m-th subpath in the c-th cluster, $\mathbf{a}_r(\theta_{c,m}^r, \phi_{c,m}^r)$ and $\mathbf{a}_t(\theta_{c,m}^t, \phi_{c,m}^t)$ are the receive and transmit array response vectors respectively, with θ and ϕ representing the azimuth and elevation angles, and τ_c is the delay of the c-th cluster. The complete parameter sets for all standardized CDL models are provided in Appendix B.

The 3GPP specifies CDL models with different DSs (CDL-A to CDL-E) and spatial characteristics for various deployment scenarios. CDL-A, CDL-B, CDL-C are constructed as NLOS scenarios, while CDL-D, CDL-E are constructed as LOS scenarios [19]. Each cluster's power follows an exponential decay profile:

$$P_c = P_0 e^{-\frac{\tau_c}{\tau_{\text{rms}}}} \tag{2.9}$$

where P_0 is the power of the first cluster, τ_{rms} is the root mean square DS, and τ_n is the cluster delay. The angular spread properties are modeled using wrapped Gaussian or Laplacian distributions, capturing the spatial correlation characteristics essential for MIMO system evaluation. These models enable realistic performance assessment of beamforming and spatial multiplexing techniques in 5G NR networks.

The channel models are particularly crucial for PUSCH and PDSCH performance evaluation, where different modulation schemes (from QPSK to 256QAM) are selected based on SNR conditions and cell configuration optimizations, including beamforming effects. The choice of SCS is primarily dictated by the deployment scenario (e.g., TDD vs. FDD), while PRB allocation depends on the required bandwidth and available resources—a key consideration in Layer 2 scheduling. The TDL and CDL models guide these decisions by simulating realistic channel conditions, enabling optimal RE utilization in both LOS and NLOS environments.

3. STATISTICAL METHODS FOR VISUALIZING HIGH-DIMENSIONAL DATA

3.1 Importance of Feature Reduction

High-dimensional datasets present significant challenges in analysis and interpretation due to the curse of dimensionality [22]. Effective feature reduction techniques are essential to transform such data into a lower-dimensional space (with $k \ll d$) while preserving intrinsic structural relationships. The primary objective is to derive an optimal nonlinear projection $f^* : \mathbb{R}^d \to \mathbb{R}^k$ (where $k < d$) that minimizes distortion of geodesic distances on the underlying manifold $\mathcal{M}$.

$$f^* = argmin_f \mathbb{E}\left[\left(\frac{d_{\mathcal{M}}(\mathbf{x}_i, \mathbf{x}_j) - \|f(\mathbf{x}_i) - f(\mathbf{x}_j)\|}{d_{\mathcal{M}}(\mathbf{x}_i, \mathbf{x}_j)}\right)^2\right] \tag{3.1}$$

where $d_{\mathcal{M}}(\cdot, \cdot)$ represents the geodesic distance between points $\mathbf{x}_i$ and $\mathbf{x}_j$ on the manifold. Feature reduction serves three critical purposes: computational efficiency by reducing dimensionality, noise mitigation through the elimination of redundant features, and interpretability by enabling visualization of complex relationships.

In wireless communication systems, high-dimensional CSI data often resides on nonlinear manifolds due to multipath propagation and hardware impairments. Traditional linear techniques like Principal Component Analysis (PCA) fail to capture these nonlinearities, necessitating advanced methods such as t-distributed Stochastic Neighbor Embedding (t-SNE) [23]. These approaches preserve local neighborhoods while revealing global patterns, facilitating the identification of clustering behavior or classification boundaries essential for LOS/NLOS classification.

The efficacy of feature reduction is particularly evident in scenarios where raw data dimensions vastly exceed the number of observations. Moreover, visualization of the reduced space provides qualitative insights into data separability and the presence of distinct operational regimes, guiding further algorithmic development.

3.2 t-Distributed Stochastic Neighbor Embedding

The t-SNE algorithm constructs a low-dimensional embedding through divergence minimization between probability distributions. The high-dimensional similarities p_{ij} are defined as symmetrized conditional probabilities:

$$p_{ij} = \frac{p_{j|i} + p_{i|j}}{2n}, \quad p_{j|i} = \frac{\exp(-\|\mathbf{x}_i - \mathbf{x}_j\|^2/2\sigma_i^2)}{\sum_{k\neq i}\exp(-\|\mathbf{x}_i - \mathbf{x}_k\|^2/2\sigma_i^2)} \tag{3.2}$$

In the low-dimensional space, the Student-t distribution with one degree of freedom models pairwise similarities:

$$q_{ij} = \frac{(1 + \|\mathbf{y}_i - \mathbf{y}_j\|^2)^{-1}}{\sum_{k\neq l}(1 + \|\mathbf{y}_k - \mathbf{y}_l\|^2)^{-1}} \tag{3.3}$$

The embedding coordinates $\{\mathbf{y}_i\}_{i=1}^n$ are obtained by minimizing the Kullback-Leibler divergence:

$$\mathcal{L} = \sum_{i\neq j} p_{ij} \log \frac{p_{ij}}{q_{ij}} \tag{3.4}$$

using gradient descent with momentum. The gradient update at iteration t is given by:

$$\mathbf{y}_i^{(t+1)} = \mathbf{y}_i^{(t)} + \eta\frac{\partial \mathcal{L}}{\partial \mathbf{y}_i} + \alpha(t)\left(\mathbf{y}_i^{(t)} - \mathbf{y}_i^{(t-1)}\right) \tag{3.5}$$

where η is the learning rate and $\alpha(t)$ the momentum term. This formulation preserves local neighborhood structures while revealing global patterns in the wireless communication dataset.

4. DEEP LEARNING

In recent years, the rapid advancement of AI has propelled DL to the forefront of data science research and applications. As a subset of ML, DL employs hierarchical NN with multiple processing layers to learn representations of data with multiple levels of abstraction. This architecture enables the automatic discovery of complex patterns and features directly from raw data, eliminating the need for manual feature engineering that dominates traditional machine learning approaches [22].

The theoretical underpinnings of DL can be traced to the universal approximation theorem, which establishes that continuous functions defined on compact subsets of $\mathbb{R}^n$ can be approximated to arbitrary precision by a neural network with just one hidden layer containing a sufficient number of units, given appropriate activation functions [24]. However, research has shown that shallow architectures typically need substantially more neurons than deep networks to achieve comparable function approximation performance [25].

4.1 Artificial Neural Network

ANNs represent the fundamental building blocks of modern DL systems. These computational models draw inspiration from biological neural networks, where interconnected processing nodes (neurons) collaboratively learn to perform tasks through exposure to data. The hierarchical organization of ANNs enables them to learn increasingly abstract features at successive layers, making them particularly suited for complex pattern recognition tasks [26].

4.1.1 Perceptron: The Fundamental Unit

The perceptron, introduced by Rosenblatt in 1958 [24], constitutes the basic computational unit of ANNs. As illustrated in Figure 4.1, each perceptron receives multiple inputs, applies learned weights, and produces an output through an activation function.

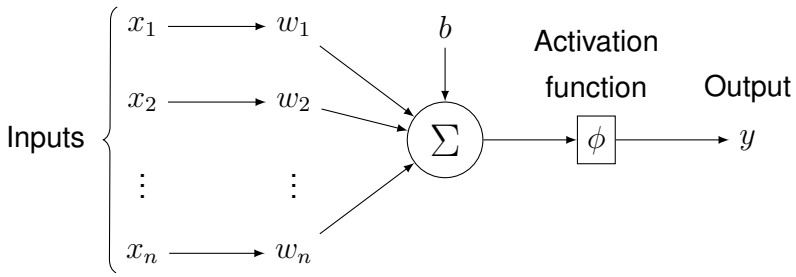


Figure 4.1. *Complete structure of an artificial neuron with activation function*

The mathematical formulation extends beyond simple weighted summation to include sophisticated operations:

$$y = \phi\left(\sum_{i=1}^{n} w_i x_i + b\right) = \phi(\mathbf{w}^T\mathbf{x} + b) \tag{4.1}$$

where $\mathbf{w} \in \mathbb{R}^n$ represents the weight vector, $b \in \mathbb{R}$ is the bias term, and $\phi(\cdot)$ denotes the activation function. The choice of activation function significantly impacts the network's learning capability:

- **Sigmoid**: $\sigma(x) = \frac{1}{1+e^{-x}}$ maps inputs to (0,1), suffering from vanishing gradients
- **Tanh**: $\tanh(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}}$ maps inputs to $(-1, 1)$, mitigating vanishing gradients compared to sigmoid but still prone to them in deep networks.
- **ReLU**: ReLU$(x) = \max(0, x)$ alleviates vanishing gradients but may cause dead neurons

The perceptron's learning mechanism follows the delta rule:

$$\Delta w_i = \eta(t - y)x_i \tag{4.2}$$

where η is the learning rate and $(t - y)$ represents the error between target and output.

4.1.2 Feedforward Neural Network Architecture

FNNs, as depicted in Figure 4.2, implement function composition through successive layers of perceptions.

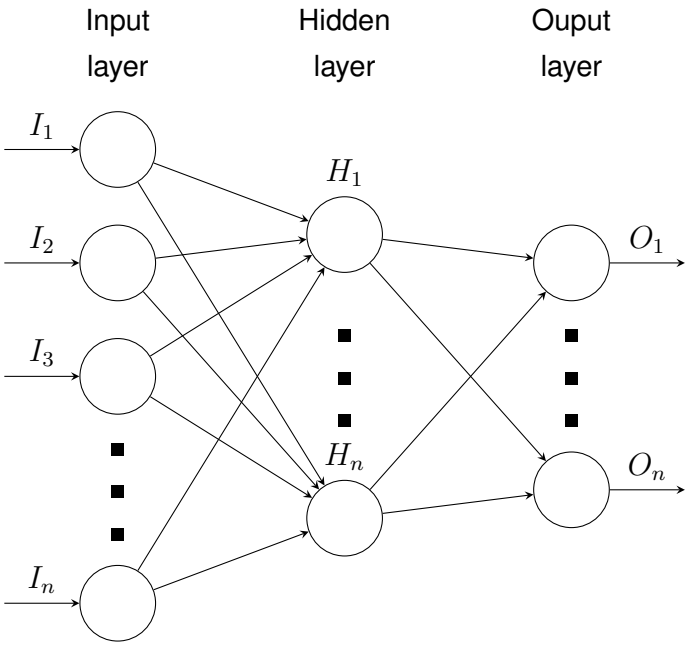


Figure 4.2. *A FNN structure is demonstrated.*

For an L-layer network, the forward propagation computes:

$$\mathbf{h}^{(l)} = \phi^{(l)}(\mathbf{W}^{(l)}\mathbf{h}^{(l-1)} + \mathbf{b}^{(l)}), \quad l = 1, ..., L \tag{4.3}$$

where $\mathbf{h}^{(0)} = \mathbf{x}$ is the input, and $\mathbf{h}^{(L)}$ produces the final output. The network's depth enables it to learn hierarchical representations:

- Lower layers capture local patterns and basic features
- Intermediate layers combine these into more complex structures
- Higher layers represent abstract concepts relevant to the task

The backpropagation algorithm [27] efficiently computes gradients through the chain rule:

$$\frac{\partial \mathcal{L}}{\partial w_{ij}^{(l)}} = \frac{\partial \mathcal{L}}{\partial z_j^{(l+1)}} \cdot \frac{\partial z_j^{(l+1)}}{\partial w_{ij}^{(l)}} \tag{4.4}$$

where $z_j^{(l)}$ represents the pre-activation of neuron j in layer l. While vanilla gradient descent uses these gradients directly for weight updates, modern optimizers employ more sophisticated approaches to accelerate convergence and improve stability.

The simplest enhancement, Stochastic Gradient Descent (SGD), introduces mini-batch sampling to reduce computational cost while adding beneficial noise to the optimization process. This noise helps escape shallow local minima early in training, though it may require careful learning rate scheduling to maintain stability in later stages.

Building upon SGD's foundation, momentum methods address its oscillatory behavior by accumulating an exponentially decaying average of past gradients:

$$v_t = \gamma v_{t-1} + \eta \nabla_\theta J(\theta) \tag{4.5}$$

This smoothing effect proves particularly valuable when navigating ravines in the loss landscape where surface curvature varies significantly across dimensions. Moreover, the momentum term γ acts as a damping factor that preserves consistent update directions while suppressing erratic oscillations orthogonal to the optimization path.

Taking this concept further, the Adaptive Moment Estimation (Adam) optimizer combines momentum with parameter-specific adaptive learning rates:

$$\theta_{t+1} = \theta_t - \frac{\eta}{\sqrt{\hat{v}_t} + \epsilon} \hat{m}_t \tag{4.6}$$

The second moment estimate v_t essentially provides an automated preconditioning that normalizes updates according to each parameter's gradient history. Notably, the inclusion of ϵ ensures numerical stability while maintaining well-defined behavior even when $\hat{v}_t$ becomes extremely small.

Recent theoretical analyses have revealed important tradeoffs between these approaches [28]. While adaptive methods like Adam demonstrate faster initial convergence, they may converge to qualitatively different solutions than momentum SGD. This occurs because their non-uniform gradient scaling across parameters alters the effective optimization trajectory [29]. Consequently, many practitioners adopt a hybrid strategy: leveraging Adam's rapid early progress before switching to momentum SGD for final convergence, particularly when seeking wide, flat minima associated with better generalization performance [30].

4.2 Convolutional Neural Network (CNN)

CNNs revolutionized computer vision by automatically learning spatial hierarchies of features. Their architecture incorporates three key principles:

1. **Local receptive fields**: Filters scan small regions of the input
2. **Shared weights**: The same filter detects patterns across all locations
3. **Spatial subsampling**: Pooling reduces dimensionality while preserving features

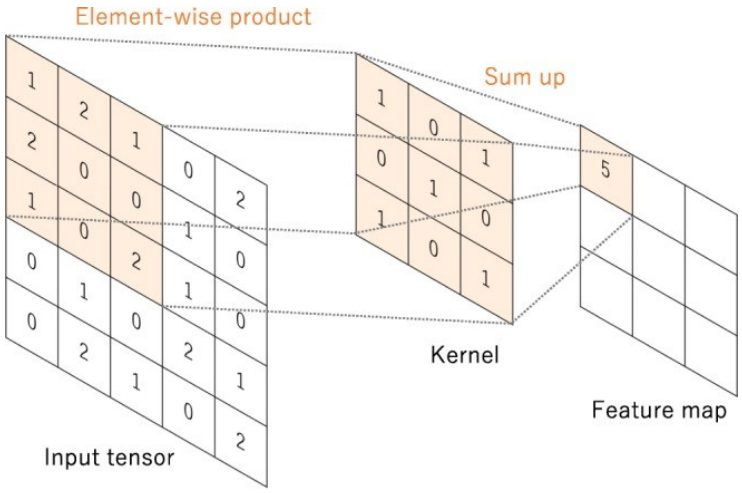


Figure 4.3. *Filtering operation is depicted in a convolutional layer, adapted from [31].*

The convolution operation (Figure 4.3) for 2D inputs extends Equation 4.7:

$$(f * g)[n, m] = \sum_{k=-\infty}^{\infty} \sum_{l=-\infty}^{\infty} f[k, l] \cdot g[n - k, m - l] \tag{4.7}$$

where f represents the input and g the filter kernel. In practice, zero-padding maintains spatial dimensions:

$$p = \frac{k - 1}{2} \tag{4.8}$$

where k is the kernel size and p the padding size.

Modern CNN architectures employ sophisticated components:

- **Batch normalization**: Normalizes layer inputs to reduce internal covariate shift

$$\hat{x}_i = \frac{x_i - \mu_B}{\sqrt{\sigma_B^2 + \epsilon}} \tag{4.9}$$

- **Skip connections**: Enable training of very deep networks via residual learning

$$\mathcal{F}(x) + x \tag{4.10}$$

- **Attention mechanisms**: Dynamically emphasize important features

$$\alpha_i = \frac{\exp(e_i)}{\sum_j \exp(e_j)} \tag{4.11}$$

In communication systems, the application of batch normalization requires careful con-

sideration distinct from its use in computer vision applications. While batch normalization has proven effective for accelerating and stabilizing deep neural network training across various domains [32], its implementation in wireless signal processing demands particular attention to signal preservation requirements.

The fundamental challenge stems from batch normalization's inherent transformation of input distributions through the operation. This redistribution of signal amplitudes and phases can prove detrimental for tasks requiring precise signal reconstruction, such as channel estimation or symbol detection, where maintaining original signal relationships is paramount [33].

For classification tasks like LOS/NLOS or channel identification, batch normalization may nevertheless offer substantial benefits despite its signal-altering properties. Contemporary research demonstrates that when the primary objective shifts to pattern recognition rather than exact signal recovery, batch normalization can enhance model convergence and generalization. Specifically, it reduces internal covariate shift during training [32], enables the use of higher learning rates without risking divergence, and provides beneficial regularization effects.

The DeepRx architecture [2] illustrates this dichotomy well, incorporating skip connections while deliberately omitting batch normalization to preserve crucial signal fidelity in channel estimation tasks. Conversely, investigations into OFDM processing [34] reveal that application of batch normalization can improve performance when the focus lies on extracting higher-level features rather than maintaining exact signal representations.

5. METHODS

5.1 Research Methodology Overview

The investigation of LOS/NLOS and channel classification in 5G NR systems follows a systematic pipeline combining theoretical analysis, simulation, and machine learning validation. As illustrated in Figure 5.1, this research adopts an integrated approach spanning three key phases: (1) generating a comprehensive dataset using standardized channel models, (2) extracting and analyzing features, and (3) designing and evaluating neural networks.

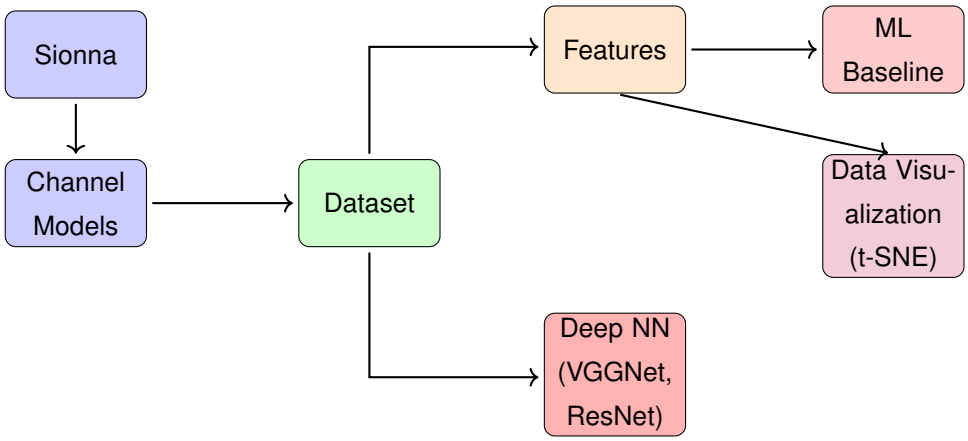


Figure 5.1. *Research workflow with two processing paths: Feature extraction for traditional ML and visualization, Direct neural network processing.*

The first phase leverages Sionna's high-fidelity 5G NR simulation capabilities to generate representative channel realizations under both LOS and NLOS conditions. This foundation ensures physical-layer accuracy while providing controlled variation across critical parameters like SNR, mobility, and antenna configuration. The generated dataset captures raw OFDM resource grids before channel estimation, preserving the complete signal characteristics needed for propagation analysis.

Building upon this dataset, the second phase develops specialized feature representations that bridge communication theory and machine learning. These features combine statistical signal properties with domain-specific transformations designed to highlight propagation condition classification. The visualization and analysis of these features,

as presented in Section 5.5.2, provide crucial insights into the inherent separability of channel conditions.

The final phase introduces novel neural architectures adapted from computer vision but specifically optimized for OFDM channel processing. The VGGNet-inspired and ResNet-inspired design incorporates antenna configuration-specific adaptations while maintaining the computational efficiency needed for practical deployment. This comprehensive methodology ensures that each component - from simulation to classification - aligns with both theoretical principles and practical implementation constraints of 5G NR systems.

5.2 Sionna

5.2.1 Introduction

The continuous evolution of 5G NR research has set the foundation for next-generation wireless technologies, with academia and industry already exploring concepts for 5G/6G networks. As identified in [12], emerging technologies like URLLC and eMBB demand advanced simulation tools capable of modeling complex physical layer interactions. Sionna emerges as a critical framework addressing these requirements through its GPU-accelerated, ML-native approach to communication system simulation.

Key research areas in next-generation wireless systems present unique challenges that Sionna specifically addresses, including spatial consistency requirements for applications like radar-based sensing and localization, where traditional stochastic channel models prove inadequate. The framework also supports the integration of NNs into L1 to be processed through its differentiable components, enabling gradient-based optimization. Furthermore, Sionna's computational efficiency makes it particularly suitable for high-fidelity simulations of MIMO systems and terahertz communications that demand substantial resources.

Developed as an open-source link-level simulator, Sionna provides researchers with a comprehensive toolkit for physical layer investigation. Its Python-based interface enables rapid prototyping of complex communication systems, allowing researchers to quickly implement and modify components through its modular design. This facilitates iterative experimentation with novel configurations as specified in [17]. The framework achieves significant performance improvements through GPU acceleration, making practical the simulation of computationally intensive scenarios like massive MIMO systems that would be prohibitive on CPU-based simulators.

5.2.2 Features

Sionna offers robust benchmarking capabilities with standardized implementations of common physical layer components. These include various Forward Error Correction schemes (LDPC, Polar, and Turbo codes), comprehensive channel models (TDL, CDL, and 3GPP 38.901 specifications), and complete OFDM processing chains. The framework's integration with TensorFlow and Keras enables seamless incorporation of ML models into traditional signal processing chains, supporting cutting-edge research in AI-native air interfaces and adaptive physical layer algorithms.

Table 5.1. *Key Components in Sionna Framework [35]*

Component Category	**Supported Features**
Channel Coding	LDPC, Polar, Convolutional, Turbo codes
Modulation	QPSK, 16QAM, 64QAM, 256QAM
Channel Models	AWGN, TDL, CDL, Ray Tracing
OFDM Processing	Resource grid, pilot patterns, equalization
Antenna Systems	Array configurations, beamforming patterns
5G NR PUSCH	Complete physical uplink shared channel implementation

A particularly noteworthy feature is Sionna's implementation of the 5G NR PUSCH, which provides researchers with a complete toolset for uplink communication analysis. This includes all necessary components for PUSCH processing, such as transport block handling, scrambling, modulation mapping, layer mapping, transform precoding, and resource element mapping, enabling comprehensive studies of 5G uplink performance under various channel conditions and system configurations.

5.3 Dataset Generation Methodology

The dataset is generated through a comprehensive L1 simulation chain that models the complete PUSCH transmission and reception process using Sionna. The generation pipeline follows these technical stages:

$$\mathcal{D} = \{(\mathbf{r}_i, \mathbf{H}_i, \mathbf{s}_i, y_i)\}_{i=1}^{N} \tag{5.1}$$

where $\mathbf{r}_i$ represents the received OFDM symbols before channel estimation, $\mathbf{H}_i$ the true CSI, $\mathbf{s}_i$ the transmitted data symbols, and $y_i \in \{0, 1\}$ the LOS/NLOS label for the i-th sample.

5.3.1 Transmitter Processing

The data generation process begins with the PUSCH transmitter chain:

$$\mathbf{s} = \Gamma(\mathbf{b}) = \text{Mod}(\text{Enc}(\text{CRC}(\mathbf{b}))) \tag{5.2}$$

where $\mathbf{b}$ is the input bit sequence, $\text{CRC}(\cdot)$ adds cyclic redundancy check bits, $\text{Enc}(\cdot)$ performs LDPC encoding, and $\text{Mod}(\cdot)$ implements the selected modulation scheme (QPSK to 256QAM). The modulated symbols are then mapped to OFDM resources:

$$\mathbf{X}[k,l] = \begin{cases} \mathbf{s}_m & \text{for data-bearing REs} \\ \mathbf{p}_n & \text{for pilot REs} \end{cases} \tag{5.3}$$

where k and l index the subcarrier and symbol respectively.

5.3.2 Channel Model Application

The OFDM signal undergoes multipath propagation through the specified channel model:

$$\mathbf{Y} = \mathbf{H} \circledast \mathbf{X} + \mathbf{n} \tag{5.4}$$

where $\circledast$ denotes circular convolution and $\mathbf{n} \sim \mathcal{CN}(0, N_0\mathbf{I})$ is complex Additive white Gaussian noise (AWGN). The channel matrix $\mathbf{H}$ is generated according to Equation 2.4 for TDL models or Equation 2.6 for CDL models.

5.3.3 Label Generation Methodology

The dataset employs two distinct labeling schemes to support different classification objectives. The first scheme provides binary labels for LOS/NLOS identification, while the second enables detailed channel model recognition through multi-class labeling.

Samples receive binary labels based on their channel model characteristics according to:

$$y_i^{\text{prop}} = \begin{cases} 0 & \text{(NLOS) for TDL-A/B/C or CDL-A/B/C models} \\ 1 & \text{(LOS) for TDL-D/E or CDL-D/E models} \end{cases} \tag{5.5}$$

This classification follows 3GPP standards where TDL/CDL variants A-C represent NLOS conditions and variants D-E represent LOS scenarios.

For detailed channel characterization, samples receive one of ten class labels:

$$y_i^{\text{channel}} = \begin{cases} m & \text{for TDL-X (A}\rightarrow\text{0, ..., E}\rightarrow\text{4)} \\ m+5 & \text{for CDL-X (A}\rightarrow\text{5, ..., E}\rightarrow\text{9)} \end{cases} \tag{5.6}$$

Label assignment occurs during dataset generation by inspecting channel configuration parameters, ensuring accurate correspondence between simulation settings and classification labels while maintaining compliance with 3GPP standards.

5.3.4 Receiver Signal Capture

The dataset captures the received signal before any channel estimation or equalization is performed:

$$\mathbf{r} = \text{Vec}(\mathbf{Y}) \tag{5.7}$$

where $\text{Vec}(\cdot)$ vectorizes the received resource grid. The final dataset consists of:

$$\mathcal{D} = \{(\mathbf{r}_i, \mathbf{H}_i, \mathbf{s}_i, y_i)\}_{i=1}^{N} \tag{5.8}$$

with each sample containing:

- Raw received symbols $\mathbf{r}_i$ (before channel estimation)
- True channel state $\mathbf{H}_i$
- Transmitted symbols $\mathbf{s}_i$
- LOS/NLOS label y_i (0 for NLOS, 1 for LOS)

5.4 Simulation Parameters Configuration

This section details the comprehensive simulation setup used to evaluate the 5G NR PUSCH performance under various channel conditions. The parameters are categorized into three groups: physical layer configuration, channel characteristics, and simulation control parameters.

5.4.1 5G NR PUSCH Parameters

The L1 configuration follows 3GPP Release 16 specifications for uplink shared channel transmission. Table 5.2 summarizes the key parameters for the OFDM-based waveform and OFDM-specific settings.

Table 5.2. *5G NR PUSCH System Configuration*

Category	**Parameter**	**Value**
System Configuration	Carrier Frequency	3.5 GHz
	Subcarrier Spacing	15 kHz
	Number of PRBs	16
	OFDM Symbols per Slot	14
	Delay Spread	300 ns
Modulation & Coding	MCS Table	2 (256-QAM)
	MCS Index	14
	Channel Coding	LDPC (BG2)
Reference Signals	DMRS Configuration	Type 1
	DMRS Symbols	[2, 11]
	DMRS Length	1 symbol
	Additional DMRS	Position 1
	CDM Groups Without Data	2
Antenna Configuration	SISO Setup	1 (BS) x 1 (UE)
	MIMO Setup	4 (BS) x 2 (UE)
	Precoding	Non-codebook (SISO)
		Codebook (MIMO)
Channel Models	TDL/CDL Variants	A, B, C (NLOS)
		D, E (LOS)

5.4.2 Channel Model and Antenna Configuration

The propagation environment is modeled using both TDL and CDL channel models, representing different multipath scenarios. The antenna configurations are specified for both SISO and MIMO cases as shown in Table 5.3.

Table 5.3. *Channel and Antenna Configuration*

Parameter	**Value**
Channel Models	
TDL Variants	A/B/C (NLOS), D/E (LOS)
CDL Variants	A/B/C (NLOS), D/E (LOS)
Number of Paths	All channel models paths
UE Speed	0-10 km/h
Carrier Frequency	3.5 GHz
Antenna Configuration	
SISO Setup	1x1 (single antenna)
MIMO Setup	2x2 (dual-polarized)
Antenna Spacing	0.5λ
Antenna Pattern	3GPP 38.901
Polarization	Vertical

5.4.3 Simulation Parameters

The evaluation methodology employs Monte Carlo simulations across a practical SNR range for 5G NR systems. Table 5.4 presents the simulation control parameters, including the number of samples and receiver algorithms used for performance analysis.

Table 5.4. *Simulation Control Parameters*

Parameter	**Value**
SNR Range	-3 dB to 18 dB
SNR Step Size	0.021 dB
Training Samples	2000 per scenario
Testing Samples	1000 per scenario (-3-18dB and specific cases at 0, 5, 10, 15 dB)
Channel Models	TDL/CDL A-E
Mobility Range	0-10 km/h

The experimental framework ensures statistically reliable results while maintaining computational efficiency. The combination of standardized channel models and practical im-

plementation parameters provides a realistic evaluation platform for 5G L1 performance analysis.

5.5 Data visualization

5.5.1 Received symbols

The constellation diagrams presented in Figures 5.2 and 5.3 provide critical insights into the signal distortion characteristics under various channel conditions and SNR levels, with red dots representing as pilots and blue dots as payloads. These visualizations capture the received symbols before any channel estimation or equalization is performed, representing the raw signal quality at the receiver input, which is used for the training purpose of NN and feature extraction. Moreover, these can be the fundamental steps for checking the validation of the dataset before applying ML techniques for classification purposes.

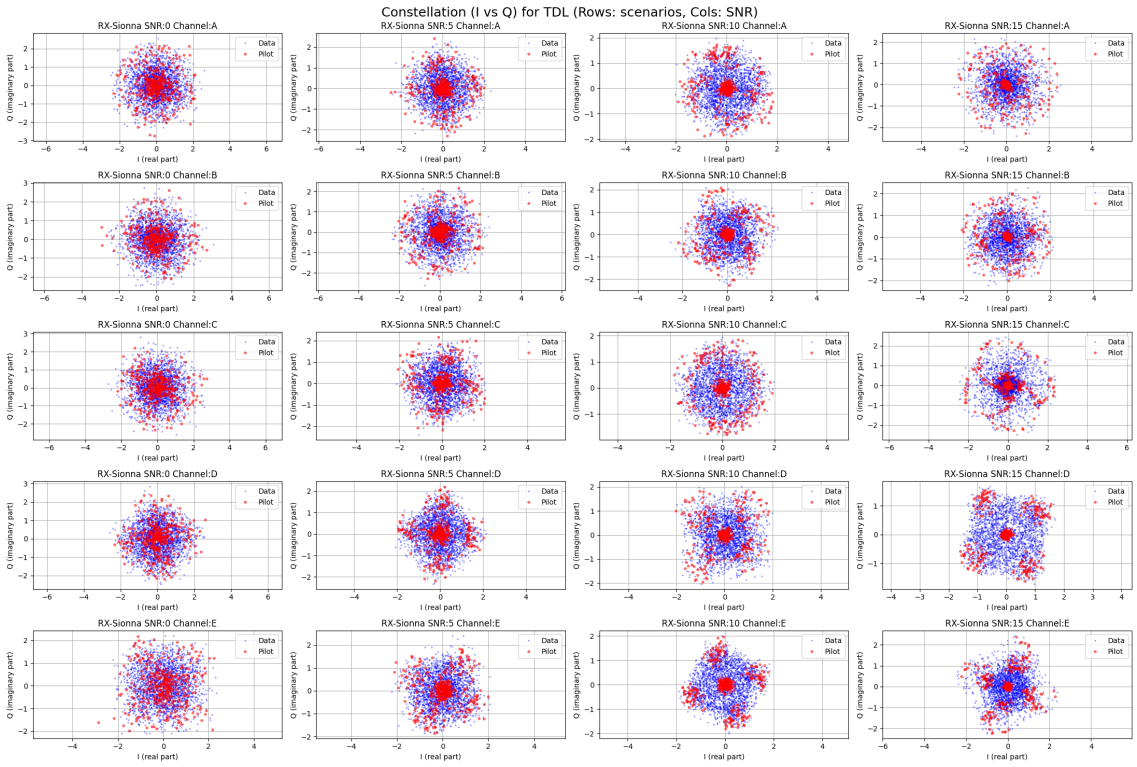


Figure 5.2. *Constellation diagrams of 1 sample for each TDL channel model A-E.*

Figure 5.2 displays the constellation patterns for TDL channel models A-E across four representative SNR values (0, 5, 10, and 15 dB). In the first observation, the concentration of pilot symbols near the constellation center stems primarily from noise dominance on severely faded subcarriers. When channel coefficients approach zero due to deep fades, the received pilots become noise-dominated, collapsing toward the origin. The central

clustering thus visually indicates subcarriers where channel estimation will face maximum uncertainty due to low effective SNR.Therefore, the higher SNR value, the less pilots concentrated in the center of the constellation.

Regarding the differences between channel models, The TDL-A (NLOS) model shows significant dispersion even at higher SNR levels, characteristic of its dense multipath profile with strong late arrivals. TDL-B demonstrates more compact clustering at 10 dB SNR compared to TDL-A, reflecting its more favorable power delay profile. The TDL-C model presents an intermediate case between A and B, with visible improvement in symbol coherence as SNR increases.

Notably, TDL-D (LOS) exhibits distinct radial spreading patterns indicative of Rician fading, where the dominant LOS component becomes clearly visible above 10 dB SNR. TDL-E shows the most concentrated constellations at high SNR, consistent with its strong LOS characteristic and limited diffuse components. The progression from 0 dB to 15 dB SNR reveals how thermal noise dominates impairment at low SNR, while channel-induced distortion becomes the primary limitation at higher SNR levels.

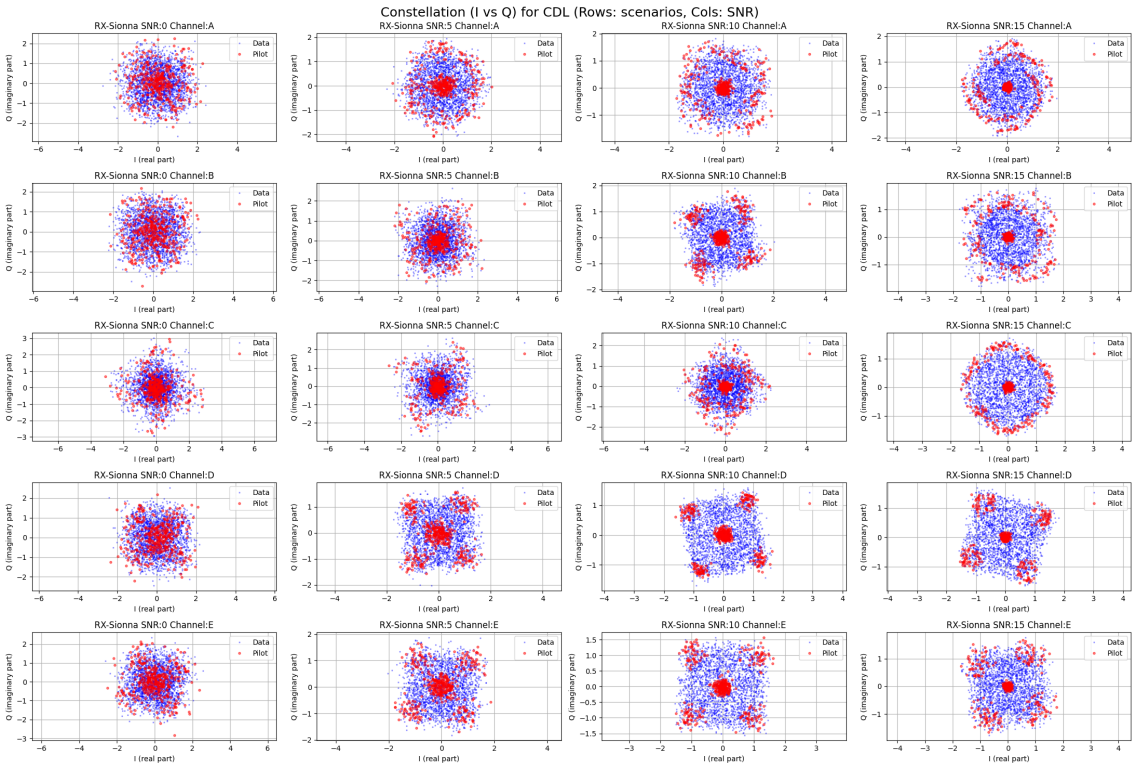


Figure 5.3. *Constellation diagrams of 1 sample for each CDL channel model A-E.*

Figure 5.3 presents the corresponding results for CDL channel models. The CDL-A (NLOS) constellation points show pronounced scattering due to its rich multipath environment with multiple clusters. CDL-B demonstrates more structured clustering compared to CDL-A, particularly evident at 15 dB SNR where the underlying 256QAM pattern be-

comes discernible. CDL-C exhibits intermediate behavior with partial symbol separation at higher SNRs.

The LOS models CDL-D and CDL-E display fundamentally different characteristics. CDL-D shows systematic displacement of constellation points due to persistent phase rotation from the dominant path, while CDL-E maintains excellent point concentration even at moderate SNR levels, reflecting its strong LOS component with minimal diffuse scattering. The angular spread characteristics of CDL models create distinct distortion patterns compared to the delay-spread dominated TDL cases.

These constellation diagrams serve two primary analytical purposes: First, they visually verify the implementation fidelity of the channel models by demonstrating expected propagation effects. Second, they provide intuitive understanding of how different channel impairments manifest in the received signal space. The visual patterns correlate directly with the expected theoretical behavior of each channel model under varying SNR conditions.

5.5.2 Data statistical visualization

The effectiveness of the selected features for distinguishing propagation conditions is evaluated through t-SNE, a nonlinear dimensionality reduction technique particularly suited for visualizing high-dimensional data. The six extracted features form a feature vector $\mathbf{f} \in \mathbb{R}^6$ for each received signal sample:

$$f_1 = \max_n(|r_n|^2) \quad \text{(Peak power)} \tag{5.9}$$

$$f_2 = \frac{\frac{1}{N}\sum_{n=1}^{N}(|r_n| - \mu)^4}{\sigma^4} - 3 \quad \text{(Excess kurtosis)} \tag{5.10}$$

$$f_3 = \frac{\frac{1}{N}\sum_{n=1}^{N}(|r_n| - \mu)^3}{\sigma^3} \quad \text{(Skewness)} \tag{5.11}$$

$$f_4 = \frac{1}{N}\sum_{n=1}^{N}\Re\{r_n\} \quad \text{(Real mean)} \tag{5.12}$$

$$f_5 = \frac{1}{N}\sum_{n=1}^{N}\Im\{r_n\} \quad \text{(Imaginary mean)} \tag{5.13}$$

$$f_6 = \sqrt{\frac{1}{N}\sum_{n=1}^{N}(|r_n| - \mu)^2} \quad \text{(Standard deviation)} \tag{5.14}$$

where $\mu = \frac{1}{N}\sum_{n=1}^{N}|r_n|$ is the mean magnitude and σ is the standard deviation of $|r_n|$.

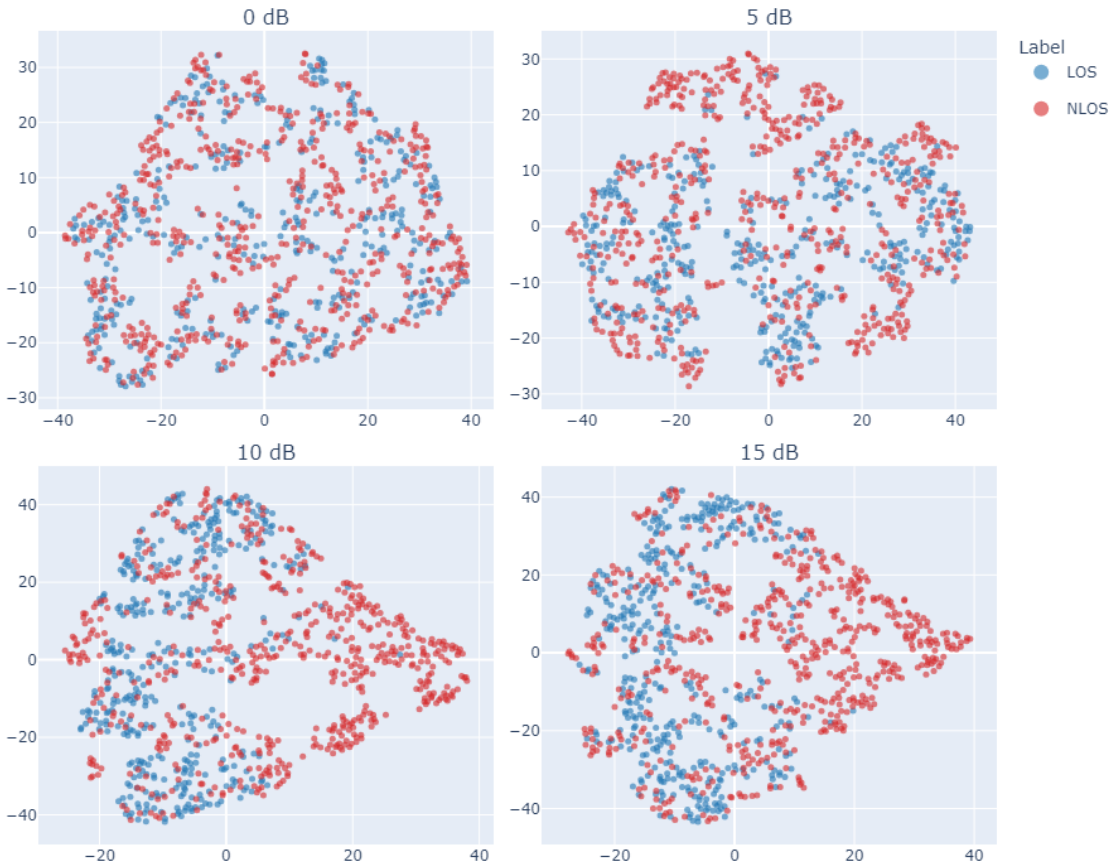


Figure 5.4. *t-SNE projection showing natural clustering of LOS (blue) and NLOS (red) conditions across all channel models. The visualization suggests reasonable separability between propagation conditions.*

Figure 5.4 reveals observable clustering tendencies between LOS and NLOS conditions with 2000 random samples of the training datasets. The LOS samples demonstrate tighter grouping patterns, while NLOS cases exhibit more dispersed distributions. This visual pattern indicates that the extracted features may naturally separate the two propagation conditions, though with some overlapping regions suggesting classification challenges near decision boundaries.

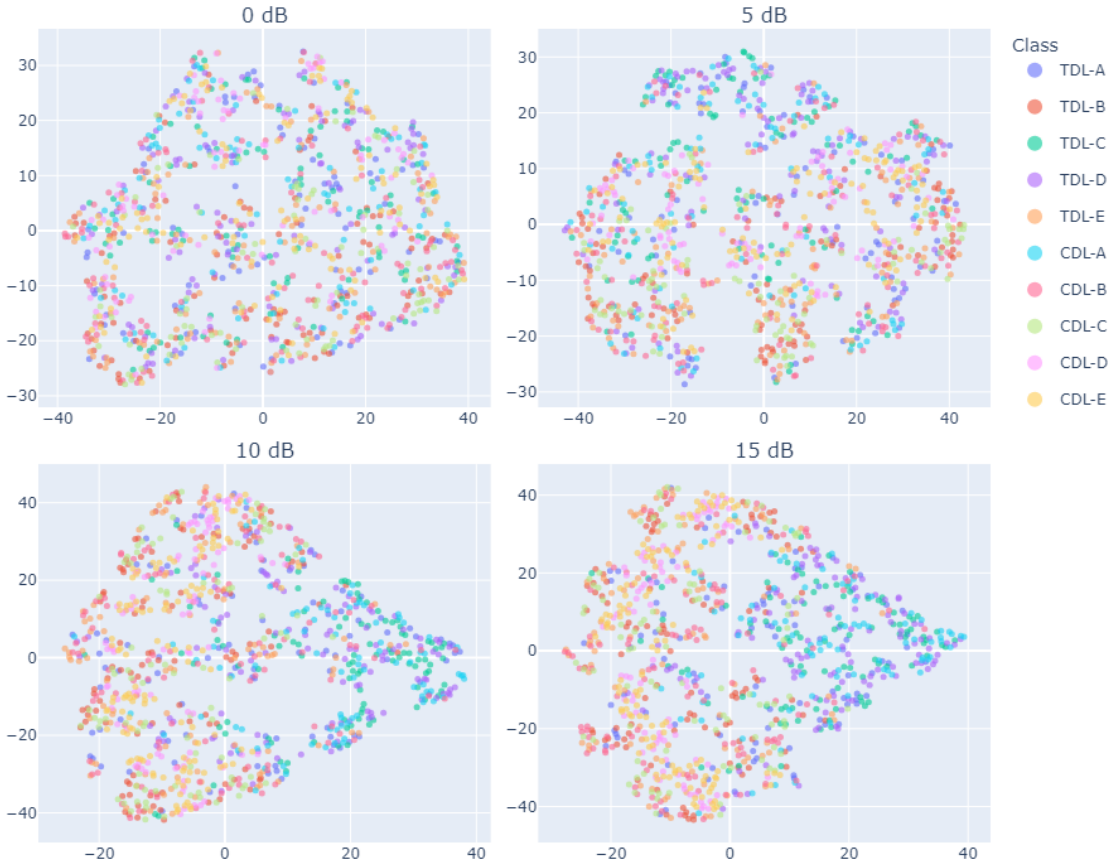


Figure 5.5. *t-SNE projection showing clustering patterns for all ten channel models (TDL A-E and CDL A-E). Colors represent different model variants, demonstrating limited model-specific separation.*

Figure 5.5 presents a more complex visualization attempting to distinguish all channel model variants. The projection shows mixed clustering patterns where model-specific groupings are less distinct compared to the LOS/NLOS separation. Some concentration patterns emerge for certain models, particularly in LOS cases, but substantial overlap exists between most classes. This suggests that while the features may capture some model-specific characteristics, the differences are less pronounced than the fundamental propagation condition distinctions.

These visualizations serve as preliminary observations of the feature space structure, suggesting that the current feature set shows potential for LOS/NLOS classification but may require enhancement for finer model classification tasks. The t-SNE projections provide an intuitive understanding of the feature space geometry before proceeding with formal classification experiments.

5.6 Neural Network Architecture

5.6.1 Multilayer Perceptron Model

The MLP serves as the baseline classifier for LOS/NLOS classification and channel classification, implemented using the feature set described in Section 5.5.2. The model architecture and training configuration are designed to provide a reliable performance benchmark while avoiding overfitting to the channel characteristics.

The network consists of an input layer with 6 neurons (matching the feature vector dimension), three hidden layers with 128, 64, and 32 neurons respectively, and an output layer with 1 neuron using sigmoid activation for binary classification. All hidden layers employ tanh activation functions, which empirically demonstrated superior performance to ReLU for this specific classification task.

The model is trained with the integration of Scikit-learn library for NLOS/LOS scenarios and Tensorflow for channel classification to ensure robustness of results and utilize the usage of GPU for multilabel training purposes:

Table 5.5. *MLP Training Configuration*

Parameter	**NLOS/LOS**	**Channel Classification**
Hidden Layers	[128, 64, 32]	[128, 64, 32]
Max Iterations	1000 epochs	1000 epochs
Activation	ReLU	Tanh
Optimizer	Adam	Adam
Batch Size	200	32
Learning Rate	0.001 (constant)	$0.001 \rightarrow 0.0001$ (exp. decay)
Early Stopping	Tolerance=1e-4	None

5.6.2 VGGNet and ResNet Architectural Principles

The VGGNet architecture [36] revolutionized DL by demonstrating the effectiveness of deep stacks of small convolutional filters. Its fundamental innovation lies in replacing large receptive fields with multiple layers of 3×3 convolutions, enabling more efficient computation of complex features through successive nonlinear transformations. This design philosophy proves particularly suitable for wireless channel analysis, where local correlations in both frequency and time domains require hierarchical feature extraction. The standard VGG architecture maintains spatial resolution through pure convolutional operations while progressively increasing channel depth, creating a feature hierarchy that

captures both local patterns and global context.

Furthermore, ResNet [37] introduced a complementary advancement through residual learning frameworks that address the degradation problem in deep networks. The fundamental innovation lies in reformulating the learning objective through skip connections that implement identity mappings:

$$\mathbf{y} = \mathcal{F}(\mathbf{x}, \{\mathbf{W}_i\}) + \mathbf{x} \tag{5.15}$$

where $\mathcal{F}$ represents the residual mapping to be learned. This architecture enables the training of substantially deeper networks while maintaining gradient flow through direct feature propagation. Figure 5.6 illustrates the residual block structure that forms the building block of ResNet architectures.

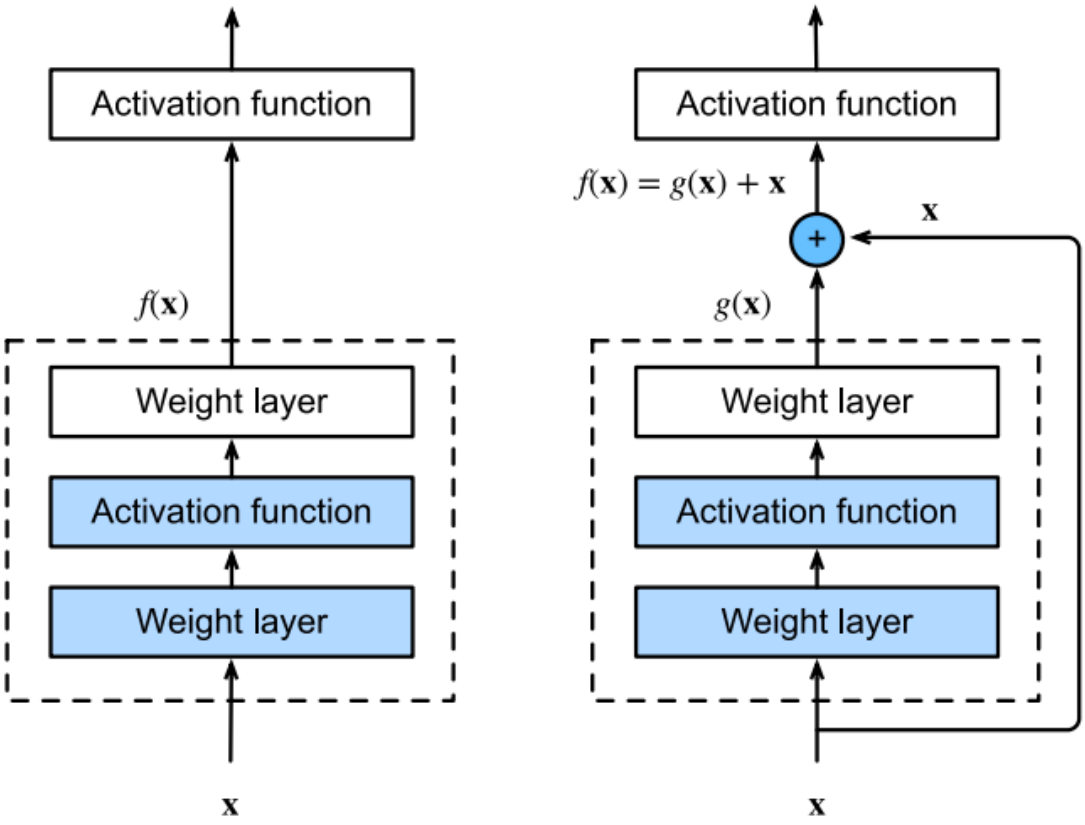


Figure 5.6. *Residual block architecture demonstrating the skip connection that enables identity mapping. The direct path preserves signal integrity while allowing gradients to flow unimpeded through deep networks.*

The integration of these architectural paradigms yields several advantages for wireless channel analysis. VGGNet's layered feature extraction provides systematic processing of OFDM resource grids, while ResNet's residual connections ensure stable training dynamics for deep networks. This combination and usage of these different model architectures

prove particularly effective for capturing both local signal distortions and global propagation characteristics essential for accurate LOS/NLOS and channel classification.

5.6.3 Complex-Valued OFDM Channel Processing

The input dimensions of 192×14 originate from the OFDM resource grid structure in 5G NR, where 192 represents the number of active subcarriers in the configuration, and 14 corresponds to the number of OFDM symbols in one time slot. This grid structure captures both frequency-domain characteristics (across subcarriers) and time-domain evolution (across symbols), providing complete channel state information for analysis. Consequently, for a received signal $\mathbf{r} \in \mathbb{C}^{192\times14}$, the input tensor construction follows:

$$\mathbf{X} = \begin{cases} [\Re(\mathbf{r}), \Im(\mathbf{r})] \in \mathbb{R}^{192\times14\times2} & \text{(SISO configuration)} \\ [\Re(\mathbf{r}_1), \Im(\mathbf{r}_1), \Re(\mathbf{r}_2), \Im(\mathbf{r}_2)] \in \mathbb{R}^{192\times14\times4} & \text{(MIMO configuration)} \end{cases} \tag{5.16}$$

The separation of complex values into real and imaginary components is motivated by three fundamental considerations in CNN processing. First, while complex-valued convolutional networks exist, they require specialized implementations and offer limited practical benefits for classification tasks, and even with backpropagation and limitations of DL training. The real-imaginary decomposition provides an efficient way to preserve phase information while using standard real-valued operations. Second, this separation maintains the linearity of complex operations when reconstructed through network layers, as the real and imaginary paths undergo identical transformations. Third, it allows the network to learn separate representations for magnitude and phase characteristics, which often exhibit different statistical properties in wireless channels.

5.6.4 VGGNet Based Model Architecture Overview

The architecture inherits VGGNet's core design principles of deep stacks of small convolutional filters while introducing several wireless-specific adaptations. The model utilizes SGD optimization with a learning rate of 0.005 and momentum of 0.9, combined with sparse categorical cross-entropy loss for training stability. This configuration provides systematic feature extraction through successive convolutional layers while maintaining manageable computational complexity for real-time applications. The design incorporates configuration-specific convolutional blocks tailored to different antenna array geometries, enabling optimal processing of both single and multiple antenna scenarios, which could be described in 5.6

Table 5.6. *Adaptive VGGNet Architecture for OFDM Processing: SISO (1×1 conv) and MIMO (3×3 conv) configurations*

Layer Type	**Output Shape**
Input	$192 \times 14 \times C$
2 x ConvNxN + ReLU	$192 \times 14 \times 32$
MaxPooling	$96 \times 7 \times 32$
2 x ConvNxN + ReLU	$96 \times 7 \times 64$
MaxPooling	$48 \times 7 \times 64$
3 x ConvNxN + ReLU	$48 \times 7 \times 128$
MaxPooling	$24 \times 7 \times 128$
3 x ConvNxN + ReLU	$24 \times 7 \times 256$
MaxPooling	$12 \times 7 \times 256$
Flatten	21504
Dense + Dropout	1024
Dense + Dropout	512
Output	2 (NLOS/LOS classification) or 10 (Channel Classification)

The single-antenna configuration employs pointwise (1×1) convolutions that operate independently on each resource element, offering several advantages for SISO scenarios. The architecture preserves exact frequency-time relationships in the OFDM grid through these localized operations, which is crucial for maintaining phase coherence across subcarriers. Furthermore, the pointwise convolutions provide efficient parameterization that focuses modeling capacity on frequency-selective fading patterns without unnecessary spatial processing overhead. From a computational perspective, this design reduces complexity significantly compared to spatial convolutions, making it suitable for real-time applications where latency constraints exist. The pointwise operations effectively implement learned frequency-domain filtering, where each 1×1 kernel develops optimal combinations of real and imaginary components specifically for propagation condition identification.

The multi-antenna configuration utilizes 3×3 spatial convolutions specifically engineered to capture antenna correlation patterns inherent in MIMO channels. These convolutional kernels operate across three adjacent subcarriers in the frequency dimension and three consecutive symbols in the time dimension, while simultaneously processing all antenna components through depth-wise operations. The spatial processing serves two primary functions in the network: it learns antenna array manifold responses through explicit mod-

eling of cross-antenna correlations, and it captures polarization diversity effects by jointly processing the real and imaginary components of each antenna's signal. The 3×3 convolution size was carefully selected to balance competing requirements - large enough to capture essential spatial context for correlation analysis, yet compact enough to maintain resolution of frequency-selective fading patterns that remain crucial for accurate channel characterization.

Both SISO and MIMO configurations maintain identical network depths, ensuring comparable levels of feature abstraction while differing only in their local processing strategies appropriate to each antenna configuration's specific requirements.

5.6.5 ResNet Based Model Architecture Overview

The ResNet architecture fundamentally differs from VGGNet through its residual connections and global average pooling layer. The skip connections enable direct gradient flow through deep networks, while global average pooling replaces fully-connected layers for more efficient spatial feature aggregation. The model employs Adam optimization with a learning rate of 0.003, providing adaptive momentum-based parameter updates. Table 5.7 will demonstrate the model architecture, with 5.7 as the BasicBlock notation with different dimension according to the output shape.

Table 5.7. *ResNet Architecture for OFDM Processing*

Layer Type	**Output Shape**
Input	$192 \times 14 \times C$
7×7 Conv, stride 2	$96 \times 7 \times 16$
BatchNorm + ReLU	$96 \times 7 \times 16$
3×3 MaxPool, stride 2	$48 \times 7 \times 16$
BasicBlock	$48 \times 7 \times 32$
BasicBlock	$24 \times 7 \times 64$
BasicBlock	$12 \times 7 \times 128$
BasicBlock	$6 \times 7 \times 256$
GlobalAveragePool	256
Dense + Dropout	512
Output	2 (NLOS/LOS classification) or 10 (Channel Classification)

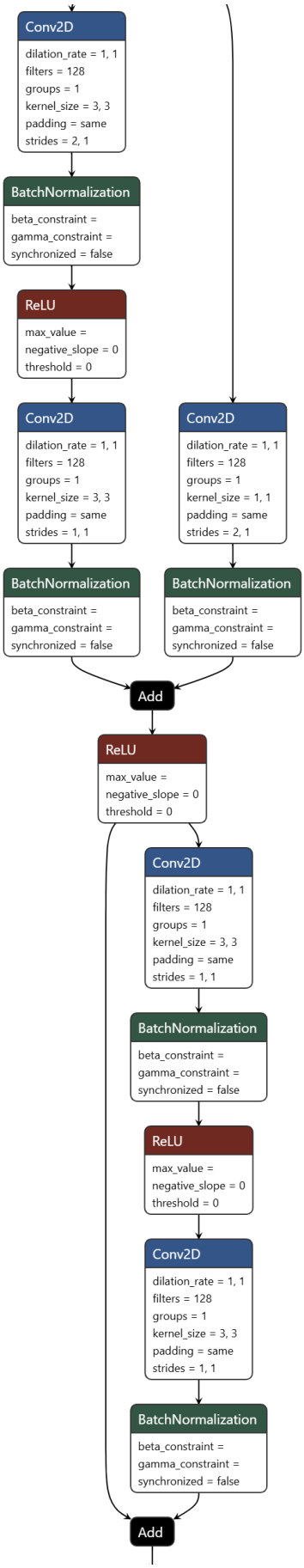


Figure 5.7. *BasicBlock demonstration of block* $12 \times 7 \times 128$ *of the Table 5.7*

The ResNet architecture's distinctive features include asymmetric pooling that preserves temporal resolution and residual blocks that maintain feature integrity across deep layers. These characteristics make it particularly suitable for analyzing OFDM signals where both local fading patterns and global channel properties contribute to accurate classification. The combination of residual learning and adaptive optimization enables more efficient training compared to conventional architectures.

The ResNet architecture introduces several innovations over VGGNet that prove particularly beneficial for wireless channel analysis. First, ResNet's use of batch normalization after each convolutional layer stabilizes training by reducing internal covariate shift, which is crucial given the varying signal statistics across different channel conditions. This contrasts with VGGNet's reliance solely on ReLU activations. Second, ResNet replaces VGGNet's flattening operation with global average pooling, significantly reducing parameters while preserving spatial information - a critical advantage for maintaining frequency-time relationships in OFDM grids. The switch from SGD to Adam optimization in ResNet provides adaptive learning rates that better handle the non-stationary gradients common in wireless channel data, particularly important when processing complex-valued signals through real-valued decompositions.

These architectural differences reflect complementary approaches to spatial feature extraction. While VGGNet's deep stacks of small convolutions excel at local pattern recognition, ResNet's residual connections enable direct propagation of both low-level signal characteristics and high-level abstractions [37]. This proves especially valuable for LOS/NLOS classification where both instantaneous signal distortions (captured by early layers) and overall propagation characteristics (learned by deeper layers) contribute to classification accuracy. The reduced parameter count from global average pooling also makes ResNet more suitable for deployment in resource-constrained edge devices common in 5G networks.

5.6.6 Task-Specific Architecture Details

The CNN architecture supports four distinct classification tasks through configuration-specific adjustments:

Table 5.8. *Architecture Parameters by Task*

Parameter	**SISO NLOS/LOS**	**MIMO NLOS/LOS**	**SISO Channel**	**MIMO Channel**
Input Channels	2	4	2	4
Kernel Size	1×1 or 3x3	3×3	1×1 or 3x3	3×3
Base Channels	16	16	16	16
Output Units	2	2	10	10
Activation	Softmax	Softmax	Softmax	Softmax

The exclusive use of 3×3 kernels in ResNet (unlike VGGNet's 1×1 for SISO) stems from architectural requirements: residual blocks inherently preserve spatial information through skip connections, making wider 3×3 convolutions essential for capturing local OFDM grid correlations while maintaining gradient flow [37]. This design choice prioritizes consistent spatial feature learning across both SISO and MIMO configurations, as 3×3 kernels better model multipath dispersion patterns regardless of antenna count, while the skip connections already provide parameter efficiency that would otherwise motivate 1×1 convolutions. For binary NLOS/LOS classification, both variants use softmax output activation with sparse categorical cross-entropy loss for training loss stability. The channel model recognition tasks employ softmax activation with 10 output units corresponding to TDL/CDL variants A-E. The MIMO configurations double the base channel counts to handle increased spatial complexity from antenna correlations.

6. RESULTS

6.1 LOS/NLOS Classification Results

6.1.1 Loss and Accuracy Analysis for NLOS/LOS classification

The training dynamics reveal fundamental differences in optimization behavior between network architectures and antenna configurations. As shown in Figures 6.1 and 6.2, the loss curves demonstrate distinct convergence patterns that reflect both architectural choices and optimization strategies. The VGGNet implementation employs SGD with momentum (0.9) and constant learning rate (0.005), while ResNet utilizes Adam optimization (initial learning rate 0.003). This fundamental difference in optimization approaches yields observable effects on the training trajectories.

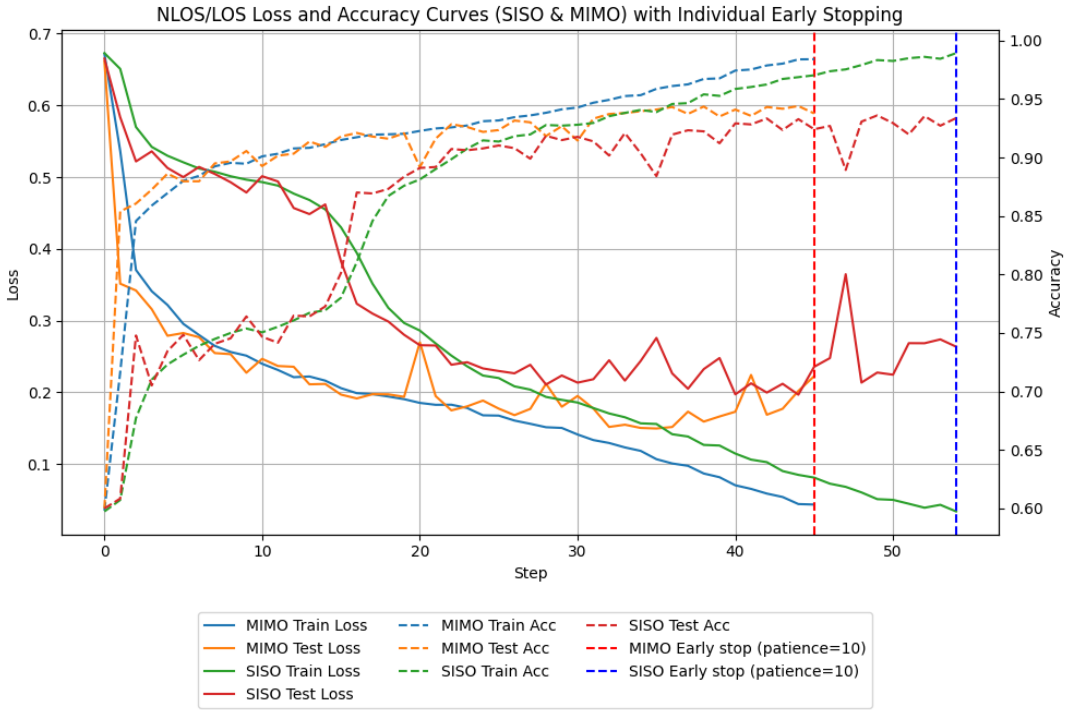


Figure 6.1. *VGGNet (SGD) training losses and accuracy for LOS/NLOS classification.*

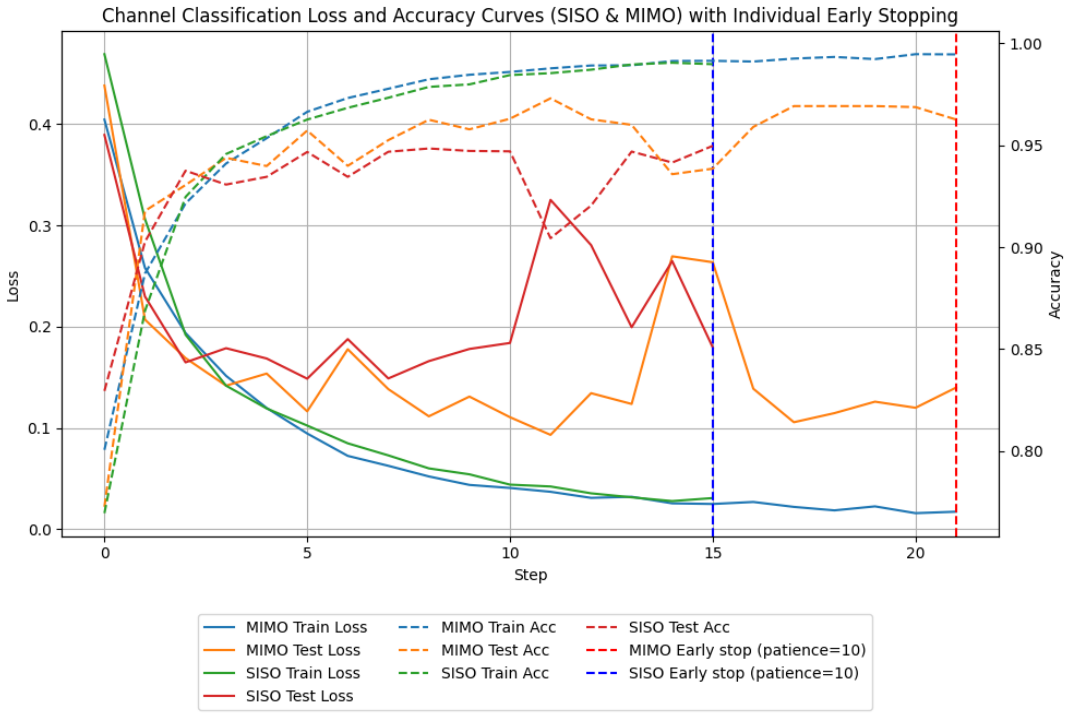


Figure 6.2. *ResNet (Adam) training losses and accuracy for LOS/NLOS classification.*

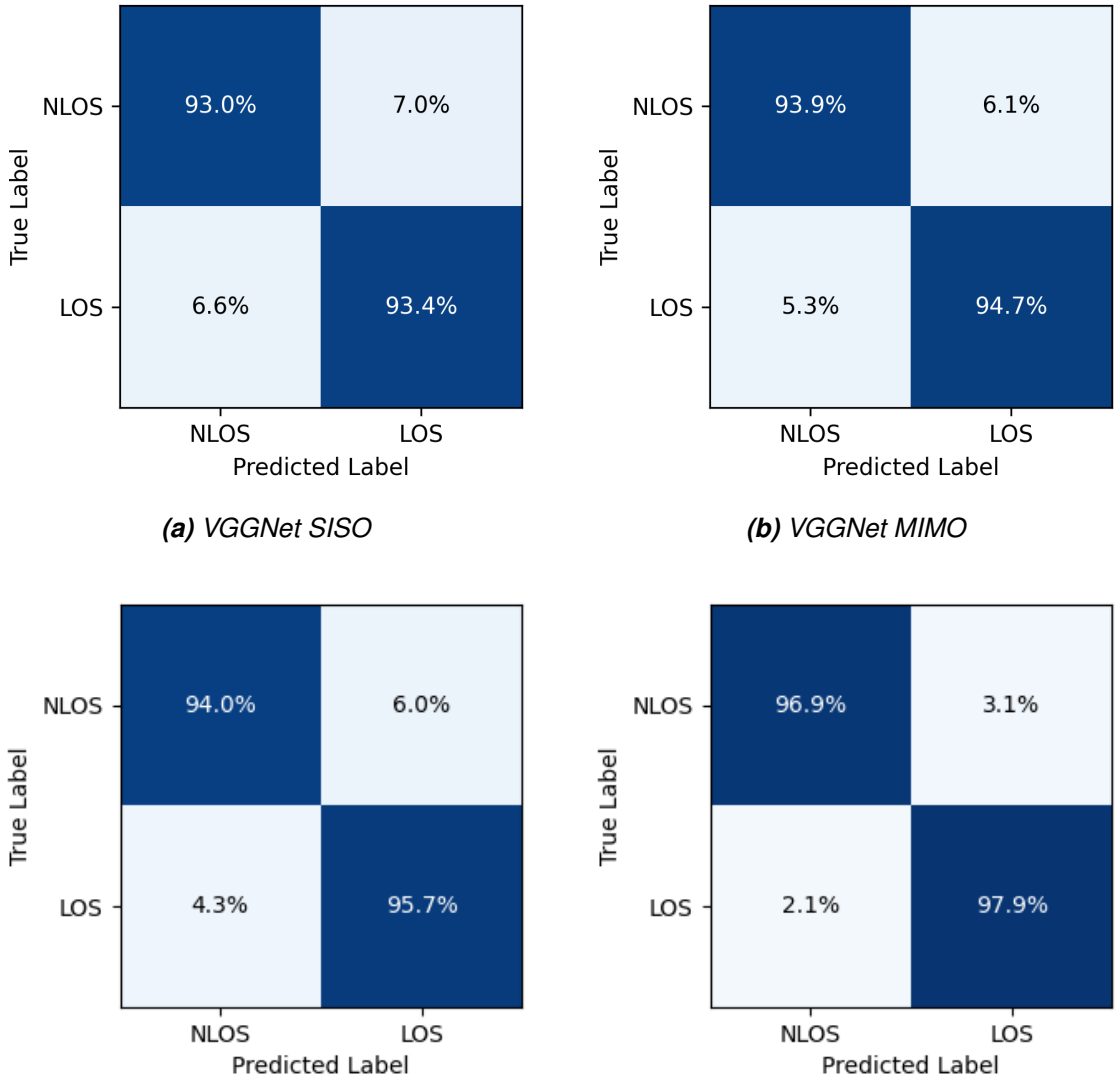
The MIMO configurations consistently achieve lower training and test loss values compared to their SISO counterparts, with ResNet showing a 25% slower convergence rate in MIMO mode. This acceleration reflects the architecture's size increase of the filter kernel for learning the spatial channel information through its residual connections. The observed increase in test loss during final epochs emerges from three concurrent factors: progressive overfitting to training samples as optimization continues, inherent stochasticity in gradient estimation during the fine-tuning phase, and the deliberate tolerance built into the early stopping mechanism to prevent premature convergence.

Three principal observations emerge from the training dynamics. First, the parallel trajectories of training and testing metrics indicate stable optimization without severe overfitting, though the MIMO configurations show slightly greater divergence in later epochs. Second, accuracy plateaus consistently appear approximately 20 epochs before loss stabilization, revealing that decision boundary formation precedes full parameter optimization. Third, the MIMO networks reach their early stopping points at higher accuracy but with marginally elevated loss values (0.15-0.2 higher), suggesting spatial processing architectures may benefit from modified stopping criteria that prioritize accuracy over loss minimization.

6.1.2 Evaluation Metrics Results for NLOS/LOS Classification

The classification performance evaluation utilizes a comprehensive testing dataset comprising 1000 samples per class (LOS and NLOS) across the full SNR range from -3 dB to 18 dB in 0.021 dB increments. This balanced evaluation set ensures statistically reliable performance measurements while maintaining realistic channel conditions representative of actual 5G NR deployment scenarios. The test samples are drawn from the same channel model distributions as the training data but represent independent realizations, with the SNR values uniformly distributed across the operational range to provide unbiased performance assessment under varying noise conditions.

The classification performance between NLOS and LOS conditions reveals important insights about the role of spatial diversity and network architecture in propagation condition identification. As shown in Figure 6.3 and Table 6.1, all configurations achieve high accuracy, but with notable architectural differences.



(a) *VGGNet SISO*

(b) *VGGNet MIMO*

(c) *ResNet SISO*

(d) *ResNet MIMO*

Figure 6.3. *Comparative confusion matrices for NLOS/LOS classification showing (a) VGGNet SISO, (b) VGGNet MIMO, (c) ResNet SISO, and (d) ResNet MIMO configurations.*

Table 6.1. *NLOS/LOS Classification Performance Metrics Comparison*

Metric	VGGNet SISO	VGGNet MIMO	ResNet SISO	ResNet MIMO
Accuracy	93.14%	94.42%	97.30%	94.67%
Precision	89.83%	93.22%	95.51%	91.40%
Recall	93.43%	92.80%	97.85%	95.67%
F1 Score	91.59%	93.01%	96.67%	93.49%

The results demonstrate that spatial diversity provides measurable but modest benefits

for NLOS/LOS classification, with VGGNet MIMO showing 1.28 percentage points higher accuracy than its SISO counterpart. More significantly, the ResNet architecture achieves identical 97.30% accuracy in both configurations, suggesting its residual connections effectively compensate for the absence of spatial diversity in SISO cases. This 2.88-4.16 percentage point improvement over VGGNet, while not dramatic, demonstrates ResNet's superior ability to learn discriminative features for propagation condition identification.

The confusion matrices reveal that ResNet maintains more balanced error rates between conditions, particularly in MIMO configuration where it achieves 97.9% LOS detection accuracy with only 2.1% false positives. This balanced performance stems from ResNet's ability to preserve gradient flow through deep networks, allowing it to learn both local signal distortions and global propagation characteristics more effectively than VGGNet's pure convolutional approach.

6.1.3 SNR-Dependent Performance for NLOS/LOS Classification

The classification accuracy under varying SNR conditions was evaluated for five ML approaches, each implemented with optimized parameters.

Table 6.2. *Algorithm Configurations for NLOS/LOS Classification*

Method	**Configuration Parameters**
kNN	Number of neighbors=5, Euclidean distance metric
SVM	RBF kernel, γ=0.1, C=1.0
Random Forest	50 estimators, max depth=10
MLP	Architecture as in Table 5.5
CNN	Architecture as in Table 5.6

The parameter configurations were optimized through grid search validation: For SVM, the RBF kernel's $\gamma = 0.1$ controls the influence radius of support vectors, while $C = 1.0$ determines the trade-off between margin maximization and classification error. The kNN's neighbor count (k=5) was selected to balance noise sensitivity and local feature capture. Random Forest uses 50 decorrelated decision trees with maximum depth 10 to prevent overfitting. All traditional methods were implemented using scikit-learn with default random seeds for reproducibility.

Table 6.3. *NLOS/LOS Classification Accuracy Across SNR Conditions (%)*

Method	15dB	10dB	5dB	0dB	Overall
SISO Configuration					
kNN	75.62	70.85	63.17	55.04	66.17
SVM	67.28	71.43	58.92	58.92	64.14
RF	83.75	80.16	73.24	56.75	73.48
MLP	85.91	82.74	76.38	57.63	75.67
ResNet-based	95.95	95.61	95.46	93.72	95.15
VGG-based	95.32	94.67	93.25	87.41	92.66
MIMO Configuration					
kNN	77.04	73.33	66.03	53.39	67.91
SVM	68.57	72.57	60.00	60.00	64.25
RF	85.41	82.43	75.38	58.19	74.95
MLP	87.01	84.41	78.49	59.09	77.99
ResNet-based	98.57	98.33	97.71	94.03	96.93
VGG-based	96.94	96.34	95.16	89.63	94.44

The ResNet-based architecture demonstrated superior noise resilience, maintaining 94.03% accuracy at 0dB SNR in MIMO configuration - a 34.94 percentage point advantage over the MLP (59.09%). This performance stems from the residual learning framework's ability to maintain gradient flow in deep networks, combined with hierarchical feature learning that effectively filters noise through successive convolutional layers. The VGG-based architecture also showed strong performance with 89.63% accuracy at 0dB SNR, demonstrating the effectiveness of its deep stack of small convolutional filters for this classification task.

6.2 Channel Model Classification Results

6.2.1 Loss and Accuracy Analysis for Channel Classification

The channel classification task presents more complex training characteristics, as evidenced in Figures 6.4 and 6.5. The combination of multiclass classification and varied channel model characteristics introduces unique optimization challenges that differ markedly from binary NLOS/LOS classification.

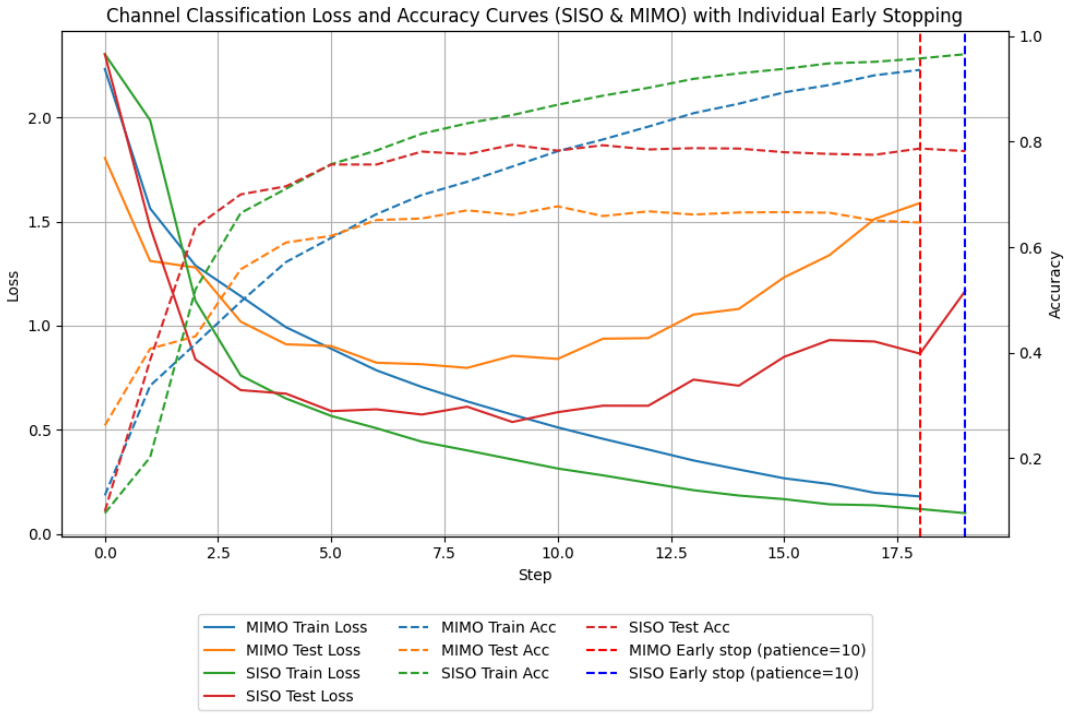


Figure 6.4. *VGGNet channel classification training and accuracy*

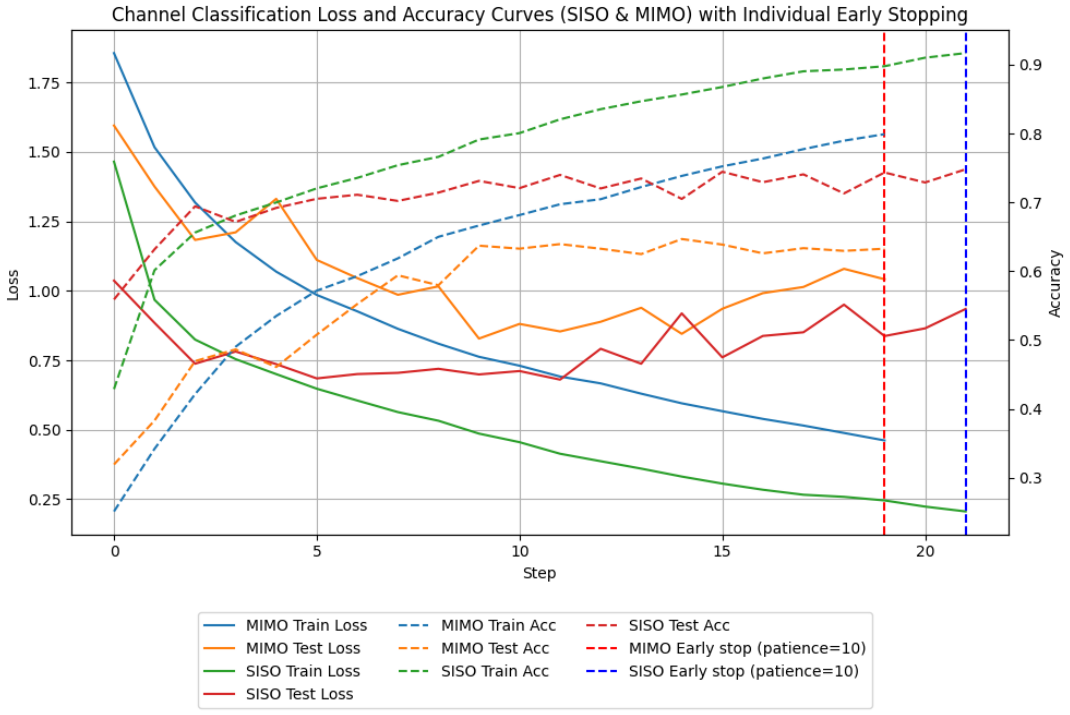


Figure 6.5. *ResNet channel classification metrics and accuracy.*

The loss curves reveal slower initial convergence compared to binary classification, with VGGNet requiring 12 epochs to reach 50% accuracy versus achieving the same milestone by epoch 5 in the NLOS/LOS task. This delayed early progress reflects the increased

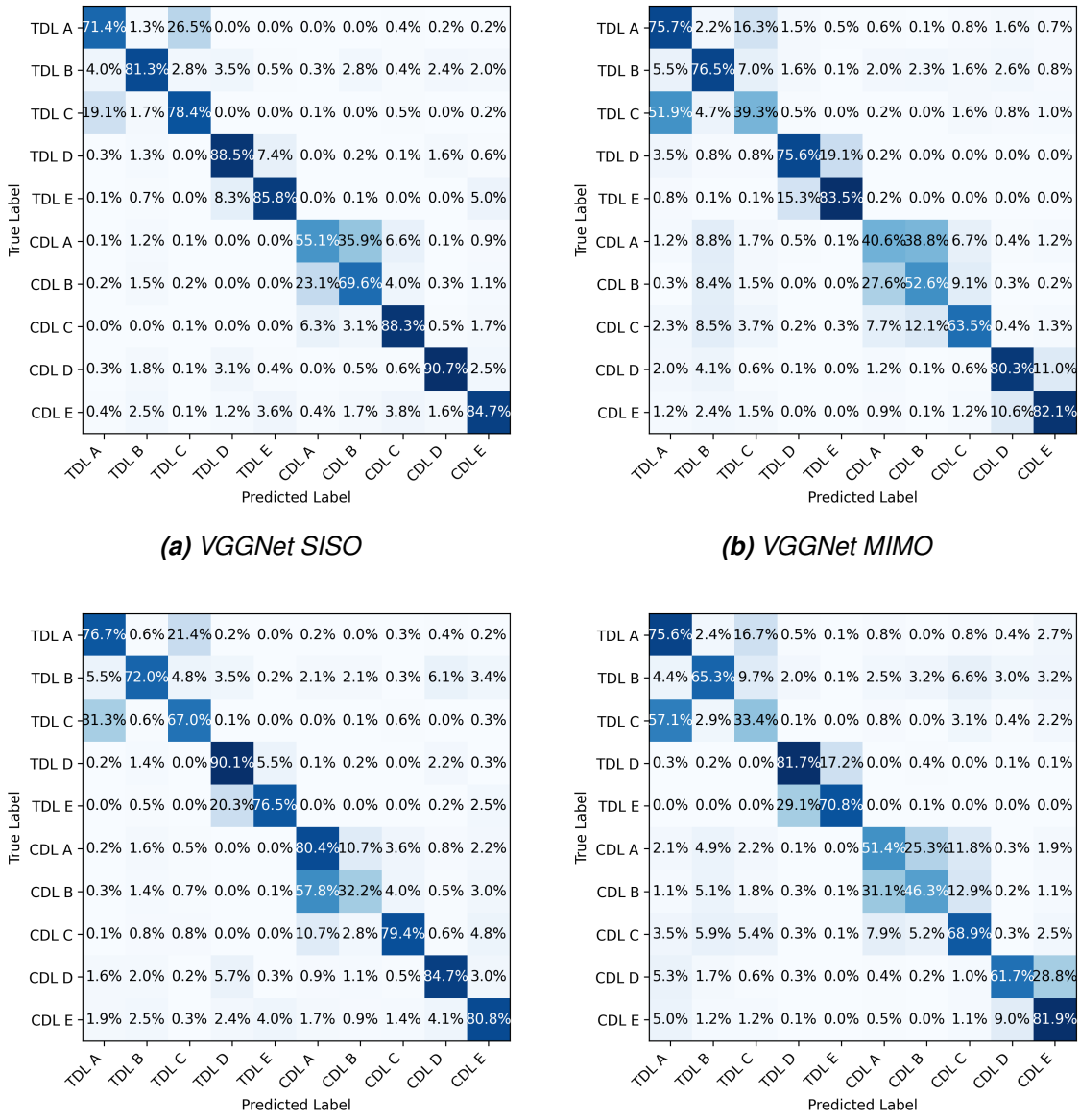
complexity of learning discriminative features across ten channel models compared to the binary propagation condition classification, where the decision boundary is fundamentally simpler to establish. The multiclass objective requires the network to simultaneously learn multiple discriminating features rather than a single decision threshold, resulting in extended initial learning phases before stable accuracy improvements emerge.

Accuracy progression exhibits several noteworthy characteristics that differ from the binary case. The SISO configuration achieves higher peak accuracy (approximately 78% vs nearly 64% for MIMO), reversing the pattern observed in NLOS/LOS classification. Both architectures show wider separation, which is a sign of increased susceptibility to overfitting, stems from the greater model complexity required for multiclass classification combined with limited training data diversity across channel models. The choice of Adam in ResNet does not demonstrate a significant difference compared to VGGNet's SGD-based optimization, likely due to adaptive learning rates amplifying noise in the gradient estimates during fine-tuning phases.

The superior final performance of VGGNet in SISO configuration (73.98% vs 71.59% accuracy) confirms that its fixed learning rate SGD optimization provides more stable convergence for this complex multiclass task. The pure convolutional architecture appears better suited for capturing the subtle temporal patterns needed for channel model classification, while ResNet's strengths in binary classification do not fully translate to this more nuanced scenario. This performance crossover highlights how optimal architecture selection depends critically on both the task complexity and the nature of the discriminative features involved.

6.2.2 Evaluation Metrics Results for Channel Classification

The channel model classification task presents surprising and counterintuitive results that reveal fundamental differences from NLOS/LOS classification. As illustrated in Figure 6.6 and quantified in Table 6.4, the performance patterns reverse significantly when moving from binary propagation condition classification to multiclass channel model identification.



(a) *VGGNet SISO*

(b) *VGGNet MIMO*

(c) *ResNet SISO*

(d) *ResNet MIMO*

Figure 6.6. *Channel model classification confusion matrices showing (a) VGGNet SISO, (b) VGGNet MIMO, (c) ResNet SISO, and (d) ResNet MIMO configurations.*

Table 6.4. *Channel Model Classification Performance Metrics Comparison*

Metric	VGGNet SISO	VGGNet MIMO	ResNet SISO	ResNet MIMO
Accuracy	73.98%	63.70%	71.59%	64.47%
Precision	75.35%	64.41%	73.96%	64.87%
Recall	73.98%	63.70%	71.59%	64.47%
F1 Score	73.57%	63.27%	71.23%	63.10%
ROC AUC	0.9643	0.9544	0.9638	0.9541

The most striking finding is VGGNet's superior performance in SISO configuration (73.98% accuracy), which outperforms both its MIMO counterpart by 10.28 percentage points and ResNet SISO by 2.39 percentage points. This reversal from the NLOS/LOS results suggests that spatial diversity actually complicates channel model classification, likely by introducing additional variability that obscures the temporal patterns critical for model identification.

The confusion matrices reveal distinct grouping patterns that correlate with propagation conditions. TDL models A-C (NLOS) show particularly high mutual confusion rates, with VGGNet SISO demonstrating the most robust classification (26.5% misclassification rate for TDL-A as TDL-C). Conversely, CDL models D-E (LOS) maintain strong performance across configurations (90.7% and 84.7% accuracy respectively in VGGNet SISO), suggesting their angular characteristics create more distinctive feature signatures.

VGGNet's advantage in this task stems from its deep stacks of small convolutional filters, which excel at capturing precise temporal patterns in the OFDM resource grid. Unlike ResNet's skip connections that help maintain gradient flow but may dilute temporal feature learning, VGGNet's pure convolutional approach provides the focused processing needed for fine-grained channel model classification. This explains why the architecture that underperforms in NLOS/LOS classification becomes superior when the task requires detailed temporal pattern recognition rather than broad propagation condition classification.

6.2.3 SNR-Dependent Performance in Channel Classification

The experimental evaluation reveals distinct performance characteristics across different SNR conditions and antenna configurations. As presented in Table 6.5, both VGG-based and ResNet-based architectures demonstrate consistent superiority over traditional machine learning approaches in both SISO and MIMO implementations, though with notable differences between models.

Table 6.5. *Channel Classification Accuracy Across SNR Conditions (%)*

Method	**15dB**	**10dB**	**5dB**	**0dB**	**Overall**
SISO Configuration					
kNN	20.76	19.79	15.82	14.92	17.82
SVM	22.74	21.08	19.83	16.60	20.06
Random Forest	27.68	25.43	22.84	21.04	24.25
MLP	26.56	24.69	21.35	21.39	23.50
ResNet-based	74.75	73.96	72.65	68.95	71.59
VGG-based	82.50	81.89	80.32	76.19	80.23
MIMO Configuration					
kNN	28.00	22.37	17.21	10.60	20.00
SVM	27.42	26.68	21.11	15.99	22.59
Random Forest	32.20	30.85	25.26	16.78	26.52
MLP	34.14	31.38	27.89	16.93	27.58
ResNet-based	76.21	73.55	64.87	54.43	64.47
VGGNet-based	76.71	74.65	67.10	59.16	66.97

In the SISO configuration, the VGGNet-based architecture achieves exceptional classification performance with an average accuracy of 80.23%, outperforming both the ResNet-based model (71.59%) and conventional methods. This represents a substantial percentage point improvement over Random Forest and other ML techniques, the best-performing traditional approach. The advantage becomes particularly pronounced in challenging low-SNR environments, where VGGNet-based maintains 76.19% accuracy at 0dB SNR compared to 68.95% for ResNet-based.

The MIMO configuration presents an interesting contrast in performance characteristics. While both DL models maintain their relative advantage over traditional methods, the VGGNet-based architecture again shows better performance (66.97% average accuracy) compared to ResNet-based (64.47%). This performance gap widens under lower SNR conditions, with the VGGNet-based achieving 59.16% accuracy at 0dB SNR compared to 54.34% for the ResNet-based model.

The observed performance patterns can be attributed to several technical factors. The additional spatial dimensions in MIMO systems introduce complexity that may obscure critical temporal and spectral features needed for accurate channel model classification.

Furthermore, the expanded parameter space in MIMO configurations likely requires more extensive training datasets to achieve optimal performance. Unlike NLOS/LOS classification, which benefits from spatial diversity, channel model classification depends more on precise identification of temporal patterns that may be better isolated in SISO configurations.

These findings carry significant implications for practical system design. For applications where channel model identification is the primary objective, SISO configurations may offer superior performance despite the general advantages of MIMO systems. The consistent outperformance of CNN architectures across all configurations suggests they should be the preferred approach for channel identification tasks. System designers should carefully consider the trade-offs between spatial diversity and feature classification when selecting antenna configurations, as the optimal choice depends on the specific application requirements and operating conditions.

7. CONCLUSION

7.1 Summary of Findings

This research has established that DL approaches can effectively classify wireless channel conditions using raw received signals as input data. The CNN architecture demonstrated robust performance across both binary NLOS/LOS classification and multiclass channel model identification tasks, validating several key hypotheses. First, the results confirm that raw OFDM symbols before channel estimation contain sufficient discriminative features for accurate propagation environment characterization. Second, the systematic variation in classification accuracy across different channel models proves that environmental changes can be reliably detected through learned parameter variations in the neural network.

The experimental outcomes reveal particularly insightful patterns regarding antenna configurations. While MIMO processing provided small advantages for binary NLOS/LOS classification, the SISO configuration surprisingly outperformed MIMO in channel model identification. This counterintuitive result suggests that spatial diversity, while beneficial for simple propagation condition detection, may introduce unnecessary complexity when classifying between detailed channel characteristics.

7.2 Future Research Directions

Several promising avenues exist for extending this work. The development of complex-valued CNN architectures represents a particularly compelling direction, as such models could more naturally process the complex baseband signals while better preserving phase relationships. This approach might yield improved performance, especially for MIMO configurations where current architectures show limitations.

The integration of attention mechanisms warrants investigation, as these could enable more focused analysis of relevant time-frequency regions within the OFDM resource grid. Such an enhancement might prove particularly valuable for distinguishing between similar channel variants that currently show high confusion rates.

Multi-task learning frameworks merit exploration as they could potentially improve fea-

ture representations through joint optimization of both propagation condition and channel model classification objectives. This approach might enhance generalization across diverse scenarios while maintaining computational efficiency.

Real-world validation studies should be conducted to evaluate the models' performance with actual measurement data, addressing potential simulation-to-reality gaps. Additionally, online adaptation techniques could be developed to enable continuous learning in deployed systems, allowing for dynamic adjustment to time-varying channel characteristics.

These future directions collectively point toward more sophisticated, adaptive channel characterization systems that could significantly enhance the intelligence and flexibility of next-generation wireless networks. The foundation established in this work demonstrates the considerable potential of data-driven approaches for L1 processing in advanced communication systems.

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APPENDIX A: TDL CHANNEL PROFILE

Table A.1. *TDL-A Channel Model Parameters (NLOS)*

Tap #	Normalized Delay	Power (dB)	Fading Distribution
1	0.0000	-13.4	Rayleigh
2	0.3819	0	Rayleigh
3	0.4025	-2.2	Rayleigh
4	0.5868	-4	Rayleigh
5	0.4610	-6	Rayleigh
6	0.5375	-8.2	Rayleigh
7	0.6708	-9.9	Rayleigh
8	0.5750	-10.5	Rayleigh
9	0.7618	-7.5	Rayleigh
10	1.5375	-15.9	Rayleigh
11	1.8978	-6.6	Rayleigh
12	2.2242	-16.7	Rayleigh
13	2.1718	-12.4	Rayleigh
14	2.4942	-15.2	Rayleigh
15	2.5119	-10.8	Rayleigh
16	3.0582	-11.3	Rayleigh
17	4.0810	-12.7	Rayleigh
18	4.4579	-16.2	Rayleigh
19	4.5695	-18.3	Rayleigh
20	4.7966	-18.9	Rayleigh
21	5.0066	-16.6	Rayleigh
22	5.3043	-19.9	Rayleigh
23	9.6586	-29.7	Rayleigh

Table A.2. *TDL-B Channel Model Parameters (NLOS)*

Tap #	Normalized Delay	Power (dB)	Fading Distribution
1	0.0000	0	Rayleigh
2	0.1072	-2.2	Rayleigh
3	0.2155	-4	Rayleigh
4	0.2095	-3.2	Rayleigh
5	0.2870	-9.8	Rayleigh
6	0.2986	-1.2	Rayleigh
7	0.3752	-3.4	Rayleigh
8	0.5055	-5.2	Rayleigh
9	0.3681	-7.6	Rayleigh
10	0.3697	-3	Rayleigh
11	0.5700	-8.9	Rayleigh
12	0.5283	-9	Rayleigh
13	1.1021	-4.8	Rayleigh
14	1.2756	-5.7	Rayleigh
15	1.5474	-7.5	Rayleigh
16	1.7842	-1.9	Rayleigh
17	2.0169	-7.6	Rayleigh
18	2.8294	-12.2	Rayleigh
19	3.0219	-9.8	Rayleigh
20	3.6187	-11.4	Rayleigh
21	4.1067	-14.9	Rayleigh
22	4.2790	-9.2	Rayleigh
23	4.7834	-11.3	Rayleigh

Table A.3. *TDL-C Channel Model Parameters (NLOS)*

Tap #	Normalized Delay	Power (dB)	Fading Distribution
1	0.0000	-4.4	Rayleigh
2	0.2099	-1.2	Rayleigh
3	0.2219	-3.5	Rayleigh
4	0.2329	-5.2	Rayleigh
5	0.2176	-2.5	Rayleigh
6	0.6366	0	Rayleigh
7	0.6448	-2.2	Rayleigh
8	0.6560	-3.9	Rayleigh
9	0.6584	-7.4	Rayleigh
10	0.7935	-7.1	Rayleigh
11	0.8213	-10.7	Rayleigh
12	0.9336	-11.1	Rayleigh
13	1.2285	-5.1	Rayleigh
14	1.3083	-6.8	Rayleigh
15	2.1704	-8.7	Rayleigh
16	2.7105	-13.2	Rayleigh
17	4.2589	-13.9	Rayleigh
18	4.6003	-13.9	Rayleigh
19	5.4902	-15.8	Rayleigh
20	5.6077	-17.1	Rayleigh
21	6.3065	-16	Rayleigh
22	6.6374	-15.7	Rayleigh
23	7.0427	-21.6	Rayleigh
24	8.6523	-22.8	Rayleigh

Table A.4. *TDL-D Channel Model Parameters (LOS)*

Tap #	Normalized Delay	Power (dB)	Fading Distribution
1	0.0000	-0.2	LOS path
1	0.0000	-13.5	Rayleigh
2	0.0350	-18.8	Rayleigh
3	0.6120	-21.0	Rayleigh
4	1.3630	-22.8	Rayleigh
5	1.4050	-17.9	Rayleigh
6	1.8040	-20.1	Rayleigh
7	2.5960	-21.9	Rayleigh
8	1.7750	-22.9	Rayleigh
9	4.0420	-27.8	Rayleigh
10	7.9370	-23.6	Rayleigh
11	9.4240	-24.8	Rayleigh
12	9.7080	-30.0	Rayleigh
13	12.5250	-27.7	Rayleigh

Note: The first tap follows a Ricean distribution with K-factor K1 = 13.3 dB and mean power of 0 dB.

Table A.5. *TDL-E Channel Model Parameters (LOS)*

Tap #	Normalized Delay	Power (dB)	Fading Distribution
1	0.0000	-0.03	LOS path
1	0.0000	-22.03	Rayleigh
2	0.5133	-15.8	Rayleigh
3	0.5440	-18.1	Rayleigh
4	0.5630	-19.8	Rayleigh
5	0.5440	-22.9	Rayleigh
6	0.7112	-22.4	Rayleigh
7	1.9092	-18.6	Rayleigh
8	1.9293	-20.8	Rayleigh
9	1.9589	-22.6	Rayleigh
10	2.6426	-22.3	Rayleigh
11	3.7136	-25.6	Rayleigh
12	5.4524	-20.2	Rayleigh
13	12.0034	-29.8	Rayleigh
14	20.6519	-29.2	Rayleigh

Note: The first tap follows a Ricean distribution with K-factor K1 = 22 dB and mean power of 0 dB.

Table A.6. *TDL-E Channel Model Parameters (LOS)*

Tap #	Normalized Delay	Power (dB)	Fading Distribution
1	0.0000	-0.03	LOS path
1	0.0000	-22.03	Rayleigh
2	0.5133	-15.8	Rayleigh
3	0.5440	-18.1	Rayleigh
4	0.5630	-19.8	Rayleigh
5	0.5440	-22.9	Rayleigh
6	0.7112	-22.4	Rayleigh
7	1.9092	-18.6	Rayleigh
8	1.9293	-20.8	Rayleigh
9	1.9589	-22.6	Rayleigh
10	2.6426	-22.3	Rayleigh
11	3.7136	-25.6	Rayleigh
12	5.4524	-20.2	Rayleigh
13	12.0034	-29.8	Rayleigh
14	20.6519	-29.2	Rayleigh

Note: The first tap follows a Ricean distribution with K-factor K1 = 22 dB and mean power of 0 dB. Subsequent taps follow Rayleigh distribution.

APPENDIX B: CDL CHANNEL PROFILE

Table B.1. *CDL-B Channel Model Parameters (NLOS)*

Clusters						
Cluster #	**Normalized Delay**	**Power (dB)**	**AoD (°)**	**AoA (°)**	**ZoD (°)**	**ZoA (°)**
1	0.0000	0.0	9.3	-173.3	105.8	78.9
2	0.1072	-2.2	9.3	-173.3	105.8	78.9
3	0.2155	-4.0	9.3	-173.3	105.8	78.9
4	0.2095	-3.2	-34.1	125.5	115.3	63.3
5	0.2870	-9.8	-65.4	-88.0	119.3	59.9
6	0.2986	-1.2	-11.4	155.1	103.2	67.5
7	0.3752	-3.4	-11.4	155.1	103.2	67.5
8	0.5055	-5.2	-11.4	155.1	103.2	67.5
9	0.3681	-7.6	-67.2	-89.8	118.2	82.6
10	0.3697	-3.0	52.5	132.1	102.0	66.3
11	0.5700	-8.9	-72.0	-83.6	100.4	61.6
12	0.5283	-9.0	74.3	95.3	98.3	58.0
13	1.1021	-4.8	-52.2	103.7	103.4	78.2

Per-Cluster Parameters					
Parameter	C_{ASD}	C_{ASA}	C_{ZSD}	C_{ZSA}	XPR
Unit	°	°	°	°	dB
Value	10	22	3	7	8

Table B.2. *CDL-A Channel Model Parameters (NLOS)*

Clusters						
Cluster #	**Normalized Delay**	**Power (dB)**	**AoD (°)**	**AoA (°)**	**ZoD (°)**	**ZoA (°)**
1	0.0000	-13.4	-178.1	51.3	50.2	125.4
2	0.3819	0.0	-4.2	-152.7	93.2	91.3
3	0.4025	-2.2	-4.2	-152.7	93.2	91.3
4	0.5868	-4.0	-4.2	-152.7	93.2	91.3
5	0.4610	-6.0	90.2	76.6	122.0	94.0
6	0.5375	-8.2	90.2	76.6	122.0	94.0
7	0.6708	-9.9	90.2	76.6	122.0	94.0
8	0.5750	-10.5	121.5	-1.8	150.2	47.1
9	0.7618	-7.5	-81.7	-41.9	55.2	56.0
10	1.5375	-15.9	158.4	94.2	26.4	30.1
11	1.8978	-6.6	-83.0	51.9	126.4	58.8
12	2.2242	-16.7	134.8	-115.9	171.6	26.0
13	2.1718	-12.4	-153.0	26.6	151.4	49.2
14	2.4942	-15.2	-172.0	76.6	157.2	143.1
15	2.5119	-10.8	-129.9	-7.0	47.2	117.4
16	3.0582	-11.3	-136.0	-23.0	40.4	122.7
17	4.0810	-12.7	165.4	-47.2	43.3	123.2
18	4.4579	-16.2	148.4	110.4	161.8	32.6
19	4.5695	-18.3	132.7	144.5	10.8	27.2
20	4.7966	-18.9	-118.6	155.3	16.7	15.2
21	5.0066	-16.6	-154.1	102.0	171.7	146.0
22	5.3043	-19.9	126.5	-151.8	22.7	150.7
23	9.6586	-29.7	-56.2	55.2	144.9	156.1

Per-Cluster Parameters					
Parameter	C_{ASD}	C_{ASA}	C_{ZSD}	C_{ZSA}	XPR
Unit	°	°	°	°	dB
Value	5	11	3	3	10

Table B.3. *CDL-C Channel Model Parameters (NLOS)*

Clusters						
Cluster #	**Normalized Delay**	**Power (dB)**	**AoD (°)**	**AoA (°)**	**ZoD (°)**	**ZoA (°)**
1	0.0000	-4.4	-46.6	-101.0	97.2	87.6
2	0.2099	-1.2	-22.8	120.0	98.6	72.1
3	0.2219	-3.5	-22.8	120.0	98.6	72.1
4	0.2329	-5.2	-22.8	120.0	98.6	72.1
5	0.2176	-2.5	-40.7	-127.5	100.6	70.1
6	0.6366	0.0	0.3	170.4	99.2	75.3
7	0.6448	-2.2	0.3	170.4	99.2	75.3
8	0.6560	-3.9	0.3	170.4	99.2	75.3
9	0.6584	-7.4	73.1	55.4	105.2	67.4
10	0.7935	-7.1	-64.5	66.5	95.3	63.8
11	0.8213	-10.7	80.2	-48.1	106.1	71.4
12	0.9336	-11.1	-97.1	46.9	93.5	60.5
13	1.2285	-5.1	-55.3	68.1	103.7	90.6

Per-Cluster Parameters					
Parameter	C_{ASD}	C_{ASA}	C_{ZSD}	C_{ZSA}	XPR
Unit	°	°	°	°	dB
Value	2	15	3	7	7

Table B.4. *CDL-D Channel Model Parameters (LOS)*

Clusters							
Cluster #	**Cluster PAS**	**Norm. Delay**	**Power (dB)**	**AoD (°)**	**AoA (°)**	**ZoD (°)**	**ZoA (°)**
1	Specular (LOS path)	0	-0.2	0	-180	98.5	81.5
	Laplacian	0	-13.5	0	-180	98.5	81.5
2	Laplacian	0.035	-18.8	89.2	89.2	85.5	86.9
3	Laplacian	0.612	-21.0	89.2	89.2	85.5	86.9
4	Laplacian	1.363	-22.8	89.2	89.2	85.5	86.9
5	Laplacian	1.405	-17.9	13.0	163.0	97.5	79.4
6	Laplacian	1.804	-20.1	13.0	163.0	97.5	79.4
7	Laplacian	2.596	-21.9	13.0	163.0	97.5	79.4
8	Laplacian	1.775	-22.9	34.6	-137.0	98.5	78.2
9	Laplacian	4.042	-27.8	-64.5	74.5	88.4	73.6
10	Laplacian	7.937	-23.6	-32.9	127.7	91.3	78.3
11	Laplacian	9.424	-24.8	52.6	-119.6	103.8	87.0
12	Laplacian	9.708	-30.0	-132.1	-9.1	80.3	70.6
13	Laplacian	12.525	-27.7	77.2	-83.8	86.5	72.9

Per-Cluster Parameters					
Parameter	C_{ASD}	C_{ASA}	C_{ZSD}	C_{ZSA}	XPR
Unit	°	°	°	°	dB
Value	5	8	3	3	11

Table B.5. *CDL-E Channel Model Parameters (LOS)*

Clusters							
Cluster #	**Cluster PAS**	**Norm. Delay**	**Power (dB)**	**AoD (°)**	**AoA (°)**	**ZoD (°)**	**ZoA (°)**
1	Specular (LOS path)	0.000	-0.03	0	-180	99.6	80.4
	Laplacian	0.000	-22.03	0	-180	99.6	80.4
2	Laplacian	0.5133	-15.8	57.5	18.2	104.2	80.4
3	Laplacian	0.5440	-18.1	57.5	18.2	104.2	80.4
4	Laplacian	0.5630	-19.8	57.5	18.2	104.2	80.4
5	Laplacian	0.5440	-22.9	-20.1	101.8	99.4	80.8
6	Laplacian	0.7112	-22.4	16.2	112.9	100.8	86.3
7	Laplacian	1.9092	-18.6	9.3	-155.5	98.8	82.7
8	Laplacian	1.9293	-20.8	9.3	-155.5	98.8	82.7
9	Laplacian	1.9589	-22.6	9.3	-155.5	98.8	82.7
10	Laplacian	2.6426	-22.3	19.0	-143.3	100.8	82.9
11	Laplacian	3.7136	-25.6	32.7	-94.7	96.4	88.0
12	Laplacian	5.4524	-20.2	0.5	147.0	98.9	81.0
13	Laplacian	12.0034	-29.8	55.9	-36.2	95.6	88.6
14	Laplacian	20.6419	-29.2	57.6	-26.0	104.6	78.3

Per-Cluster Parameters					
Parameter	C_{ASD}	C_{ASA}	C_{ZSD}	C_{ZSA}	XPR
Unit	°	°	°	°	dB
Value	5	11	3	7	8